بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

LIVE ONLINE COURSE | COHORT FALL 2026

AI & Machine Learning for Civil Engineers

SESSION 06 · ARTIFICIAL NEURAL NETWORKS

Artificial Neural Networks for Engineering Prediction

From neurons to defensible predictions. How a network learns, how we validate it, and when it is worth its complexity.

DATA → MODEL → VALIDATE → DECIDE

Dr. Abdul Waris
Research Associate · Universität Siegen, Germany

Technology partner QUDELTA STUDIOS



WHERE ARE WE?

Same concrete question, a more flexible model

SESSIONS 03–05

Linear models, trees, classification

$$\hat{y} = b_0 + \Sigma\, b_j x_j$$

linear regression | random forest | MAE · RMSE · R^2

Concrete strength from mix and age, then slope stability as a class.

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TODAY · SESSION 06

An artificial neural network

8 → 32 → 16 → 1

neurons & layers · activations · training & validation

Back to concrete strength: does the extra flexibility actually help?

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Back to concrete strength: does the extra flexibility actually help?

The ANN is one more candidate model. It has to earn its place against the baselines on the same test set.

THE COURSE AT A GLANCE

Eight sessions. One engineering workflow.

PART 1 · FOUNDATIONS

01 From Engineering Problems to Python DONE
AI/ML basics, Google Colab, NumPy & pandas

02 From Raw Engineering Data to an ML-Ready Dataset DONE
EDA, data quality, outliers, train/test split, leakage

03 Your First Engineering Prediction Model DONE
Linear regression, error metrics, cross-validation

04 Nonlinear Engineering Problems: Trees & Ensembles DONE
Decision trees, Random Forest, boosting

PART 2 · PREDICTION TO DECISION

05 From Prediction to Engineering Decision: Classification DONE
Logistic regression, confusion matrix, precision & recall

06 Artificial Neural Networks TODAY
Architecture, activations, training, overfitting

07 Can We Trust Our ML Model?
Feature importance, SHAP, engineering judgement
Instructor's research area

08 From Prediction to Probability
Monte Carlo, ML surrogates, probability of failure
Instructor's research area

TODAY · SESSION 06

One regression problem, three steps

01

ANN intuition

Neurons · layers · activations · learning its own features · epoch, batch, learning rate



02

Build & train an ANN

One neuron · the (32, 16) architecture · scaling · training · validation



03

Results & judgement

MAE · RMSE · R^2 · residuals · ANN vs baselines · limits



WHAT YOU SHOULD LEAVE UNDERSTANDING

EXPLAIN	Neurons, weights, biases, layers and activations
TRAIN	Scaling, loss, iterations and parameter updates
VALIDATE	Training, validation and test data, and their roles
COMPARE	ANN predictions against simpler baselines
JUDGE	Overfitting, extrapolation risk and limitations

RECAP · START WITH AN ENGINEERING QUESTION

Can mix and age predict concrete strength?

INPUTS · X

Eight variables

cement | slag | fly ash | water | superplasticizer | coarse agg. | fine agg. | age

Mix quantities in kg/m^3, age in days.

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Learn the mapping

$X \rightarrow \hat{y}$

Linear regression, random forest, or an ANN.

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Compressive strength

42.7 MPa

One predicted number per mix.

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One predicted number per mix.

This is regression: the target is a continuous number, not a category.

RECAP · THE DATASET AND TARGET

1,030 concrete mixes from the UCI repository

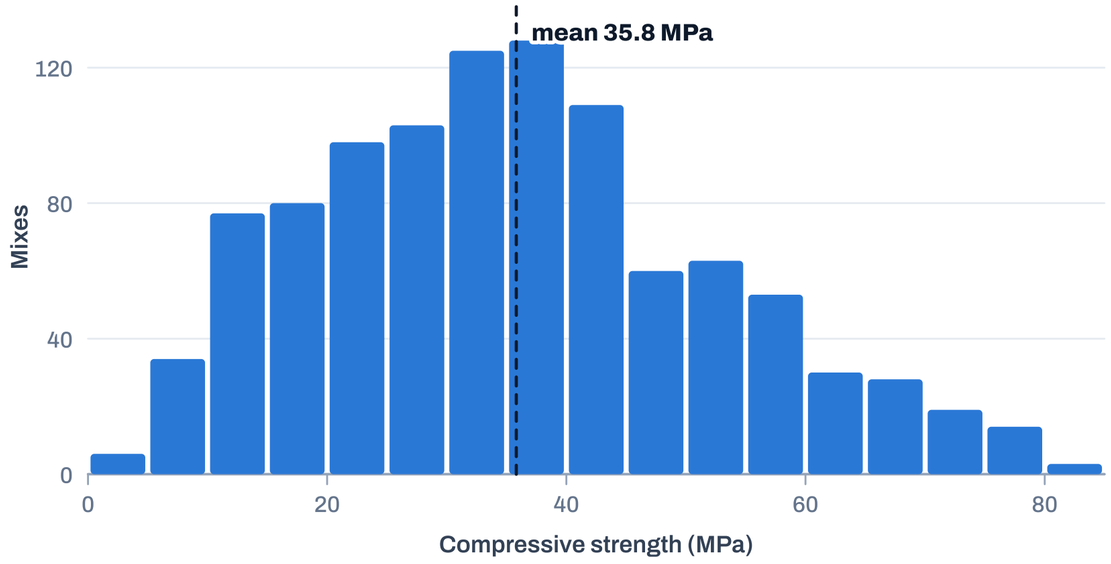
1,030
observations

8
predictor features

0
missing values

206
held-out test mixes (20%)

Strength of all 1,030 mixes



Yeh, I.-C. (1998). Concrete Compressive Strength. UCI Machine Learning Repository, CC BY 4.0. doi:10.24432/C5PK67. Original data, all 1,030 rows.

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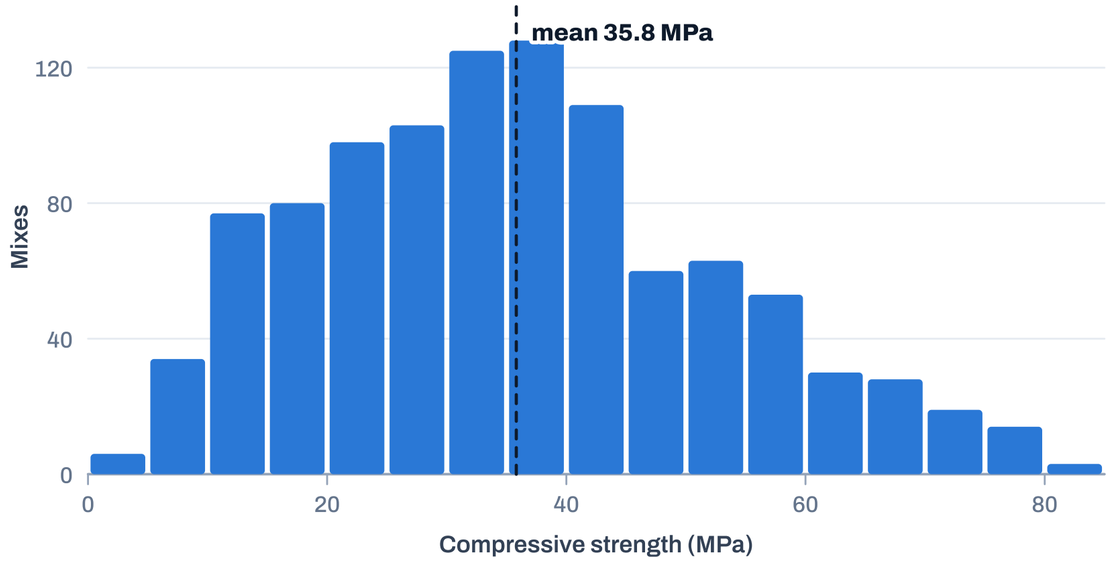
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TARGET
Compressive strength in MPa, from 2.3 to 82.6 MPa.

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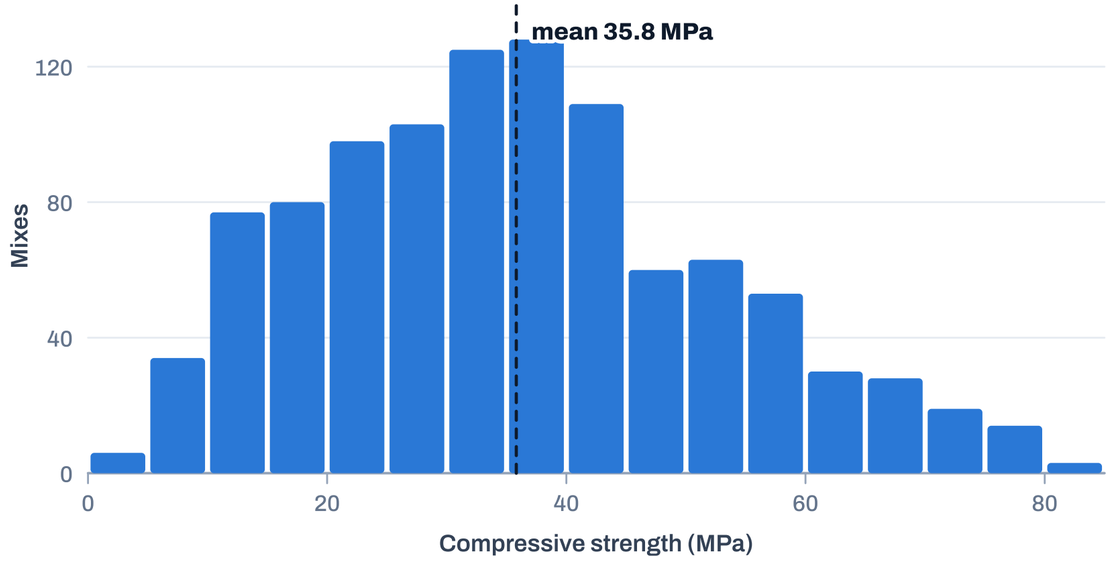
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CAUTION
The data support a prediction exercise. They are not, by themselves, a physical model of concrete behaviour.

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RECAP · WHY COMPARE MORE THAN ONE MODEL?

The ANN is one candidate, not the default winner

Multiple linear regression

A simple baseline with an additive linear form.

Session 03 · baseline

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A nonlinear ensemble of trees that captures complex patterns.

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A flexible function approximator, trained iteratively.

Today

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We compare fitted configurations on the same held-out test set. The baselines are not exhaustively tuned.

RECAP · THE BASELINE

Multiple linear regression

A REFERENCE FOR ASKING WHETHER EXTRA FLEXIBILITY IS USEFUL

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_8 x_8$$

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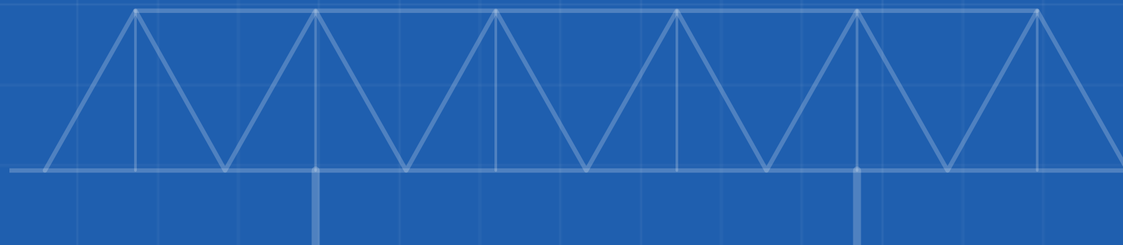
On today’s 206 test mixes the baseline reaches $R^2 = 0.628$ (RMSE 9.80 MPa). That is the bar to beat.

01

- a Where neural networks came from
- b From a brain neuron to a simple model
- c Demand prediction: layers and activations
- d A network learns its own features
- e Multilayer perceptrons
- f Epoch, batch size, learning rate

DONE WITH THE RECAP · PART 01 · ANN INTUITION

Here comes the ANN…



ARTIFICIAL NEURAL NETWORK INTUITION

Where neural networks came from

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Origin

Algorithms that try to mimic the brain

The first motivation was biological.

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Widely used, then out of favour

Fell out of favour in the late 1990s.

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More data, more computing power, better training methods.

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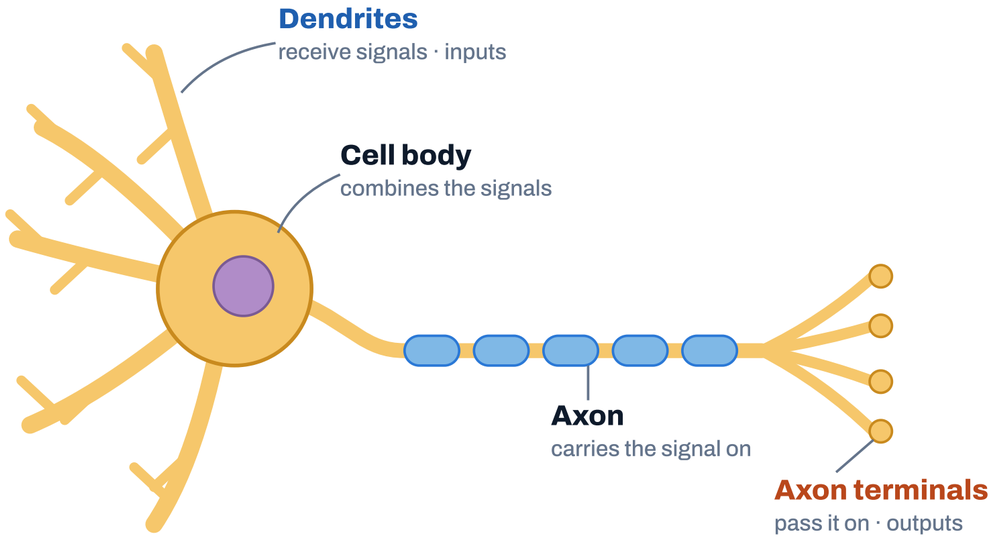
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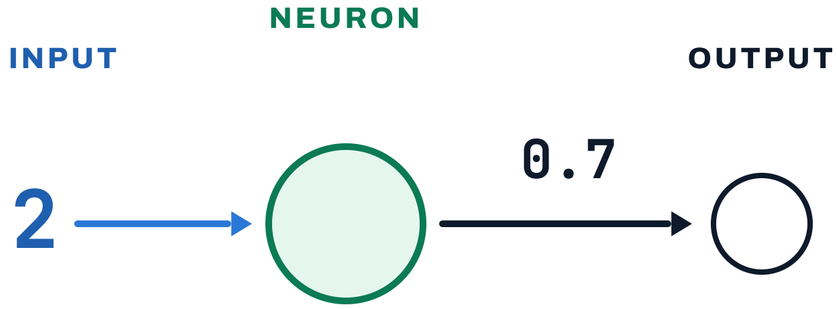
Today’s neural networks have almost nothing to do with how the brain learns. Mimicking the brain was only the early motivation.

ARTIFICIAL NEURAL NETWORK INTUITION

From a brain neuron to a simple model



SIMPLIFIED MATHEMATICAL MODEL OF A NEURON

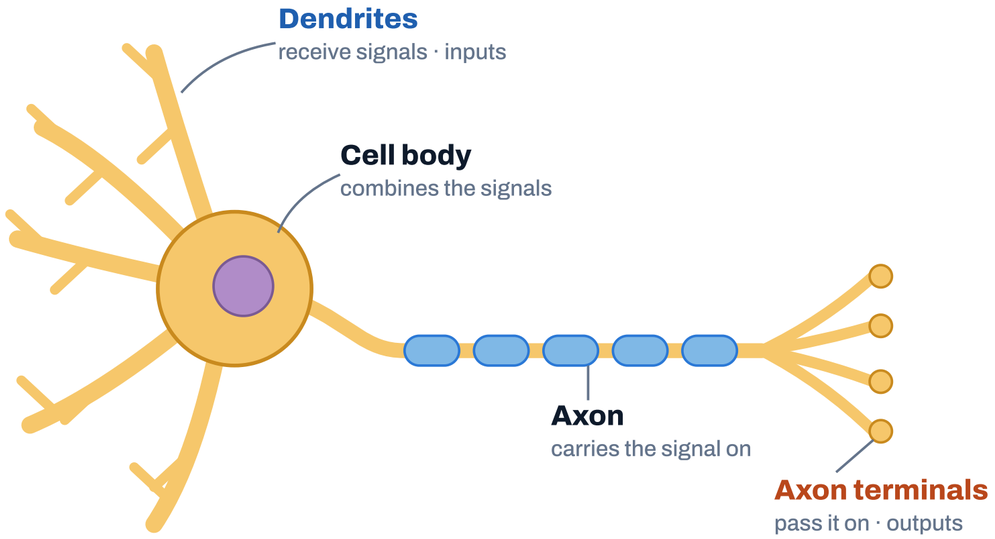


Neuron drawn schematically after the standard textbook diagram.

ARTIFICIAL NEURAL NETWORK INTUITION

From a brain neuron to a simple model

So how does the brain work? Here is a diagram of what a neuron in the brain looks like.



SIMPLIFIED MATHEMATICAL MODEL OF A NEURON

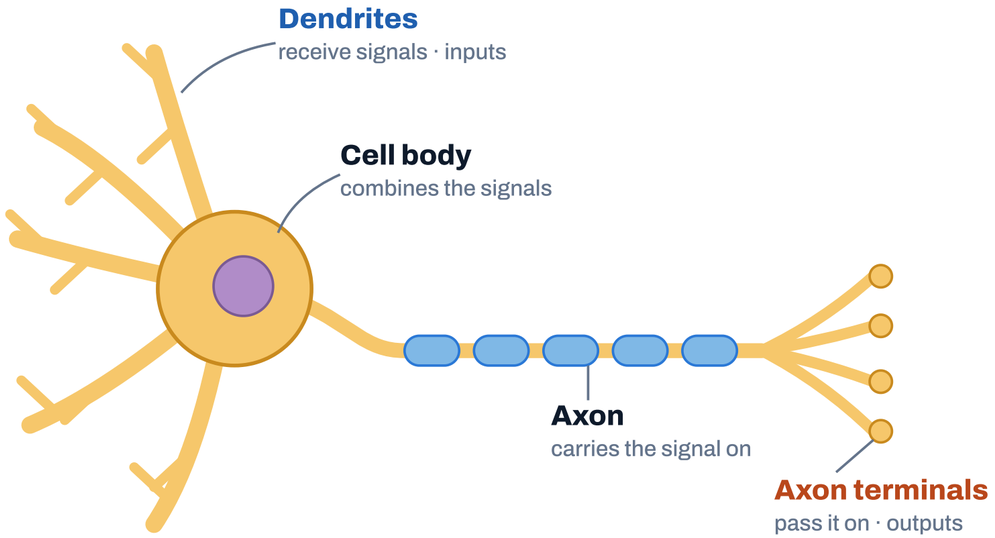


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SIMPLIFIED MATHEMATICAL MODEL OF A NEURON



With these **extremely simplified models** of a neuron we can build really powerful deep-learning algorithms. The origins were biological, but **don't take the biological motivation too seriously.**

Neuron drawn schematically after the standard textbook diagram.

DEMAND PREDICTION · WILL THIS T-SHIRT BE A TOP SELLER?

Layers and activations, one step at a time

Price

Shipping cost

Marketing

Material

Illustration adapted from Prof. Andrew Ng's lecture (Stanford Online and DeepLearning.AI).

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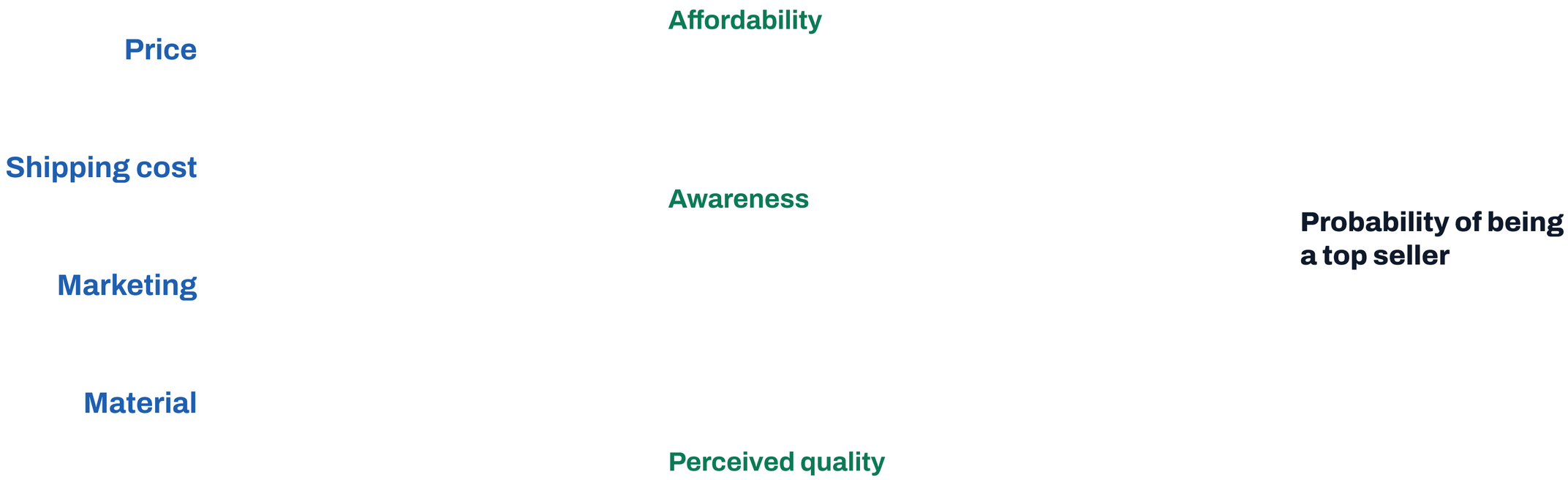


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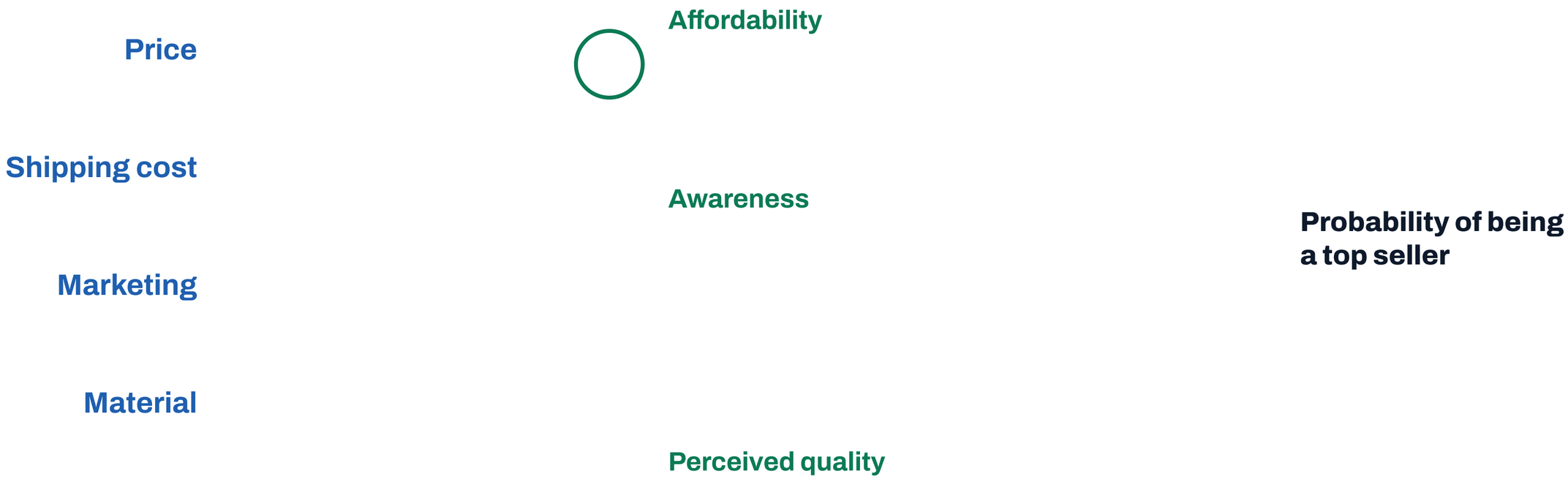


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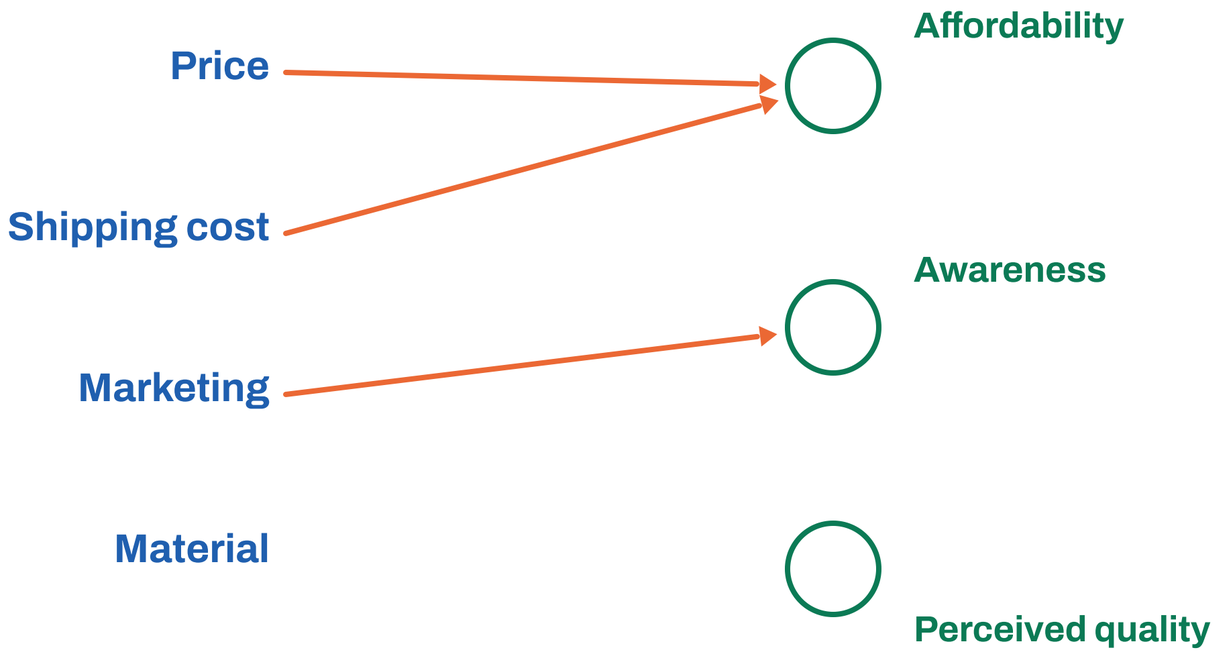
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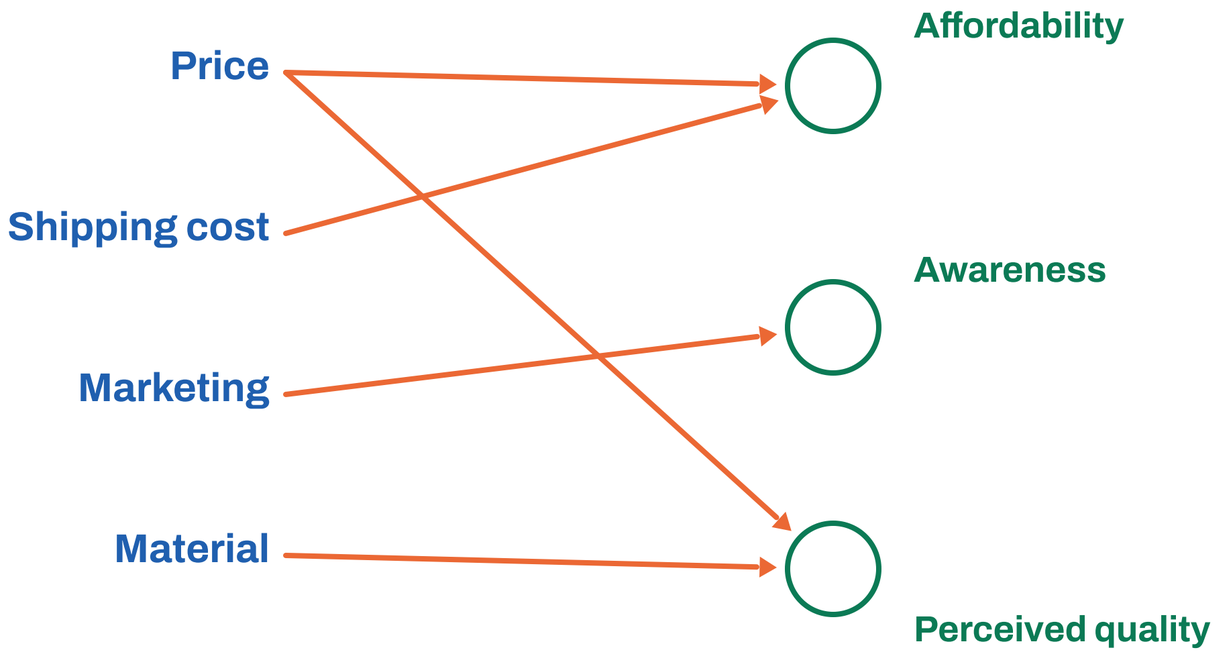


Probability of being a top seller

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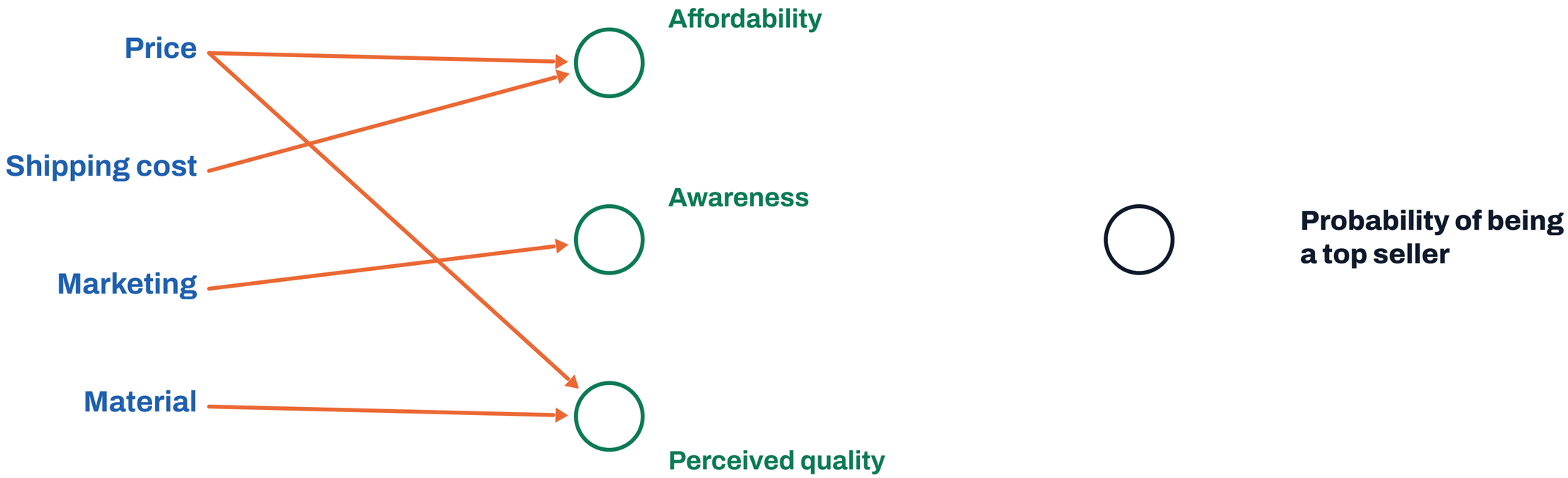


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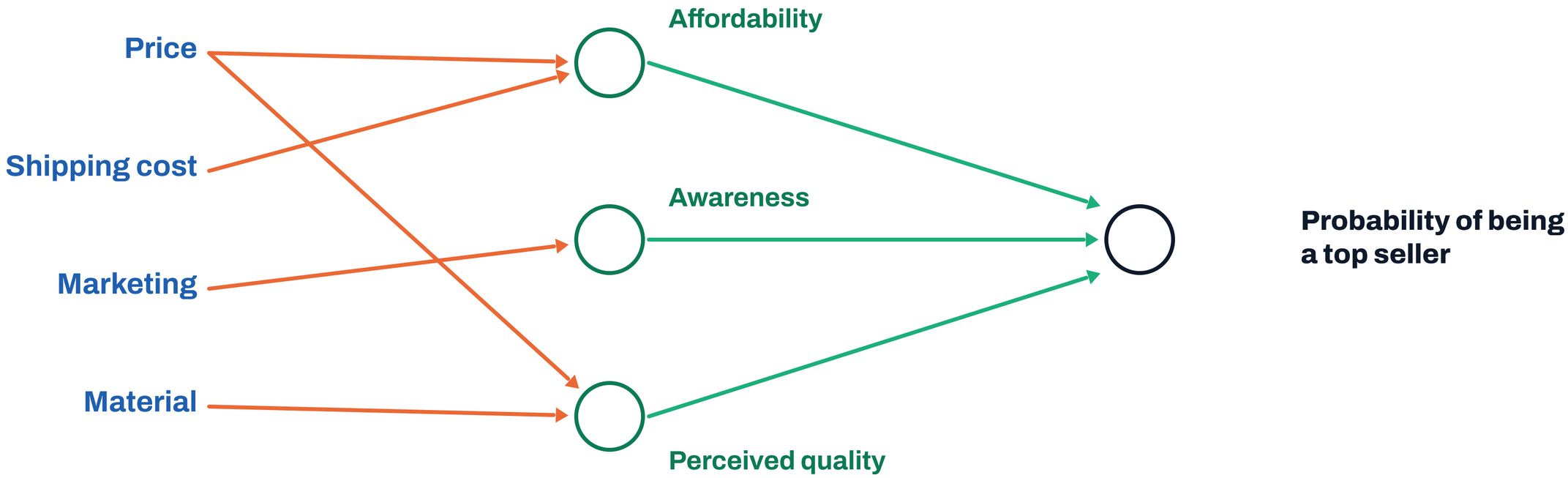


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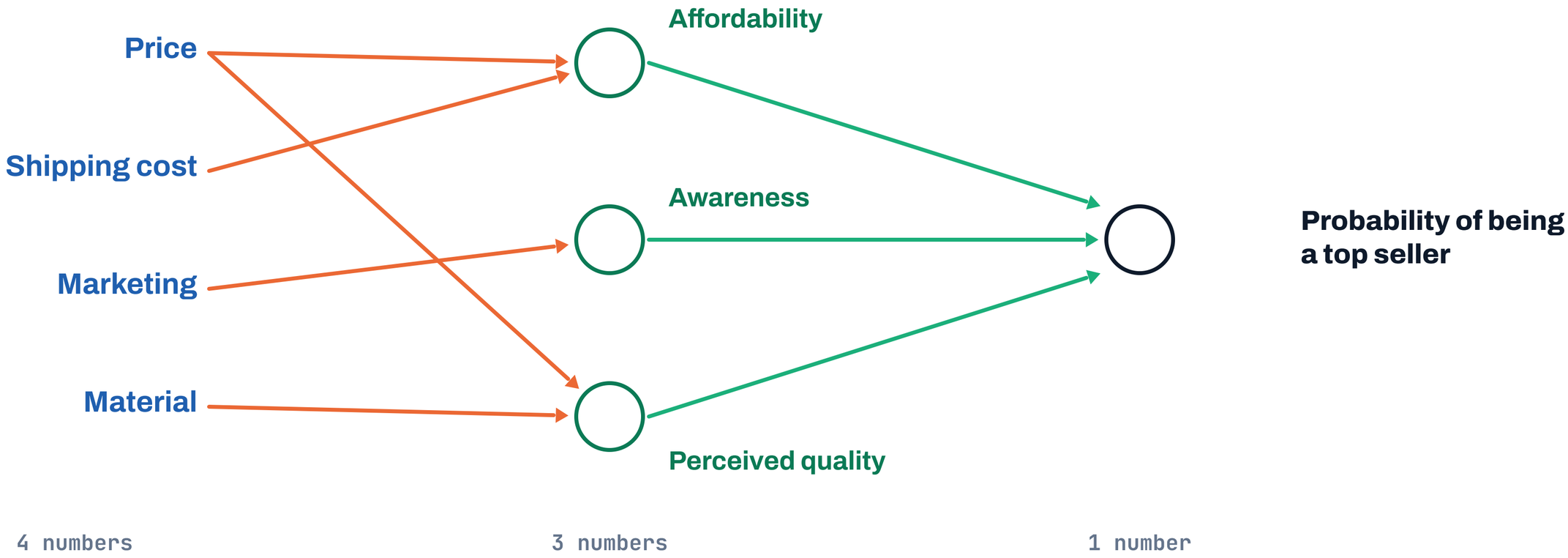


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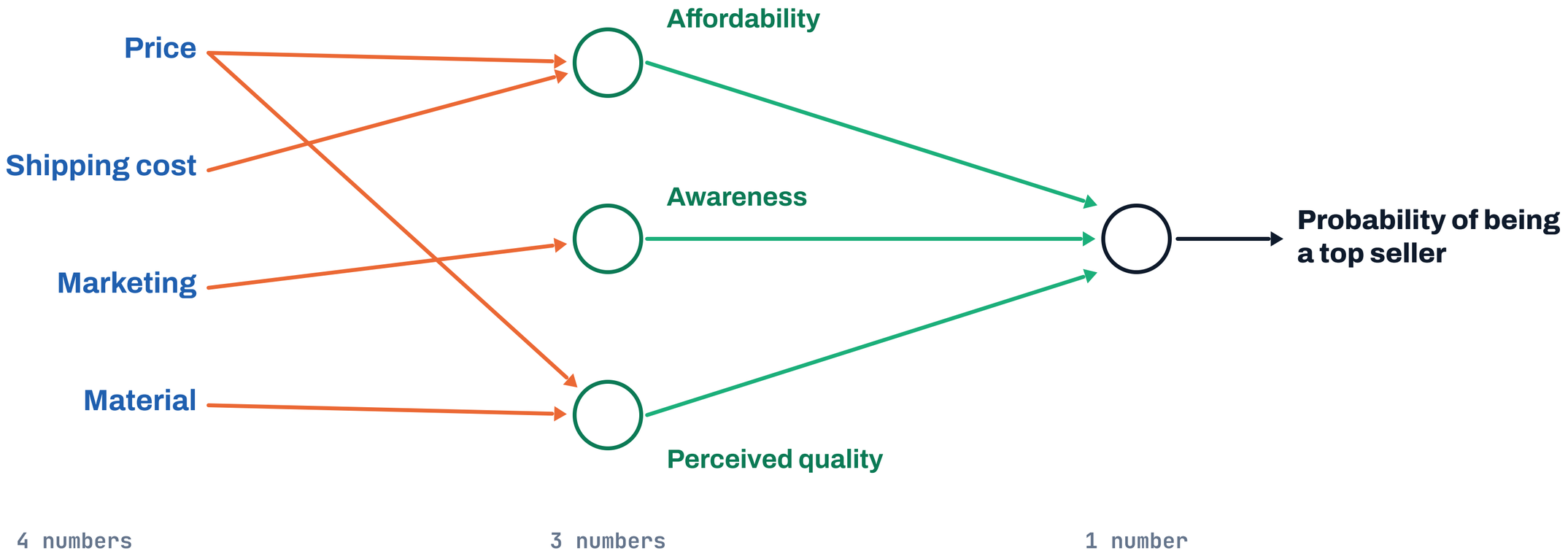


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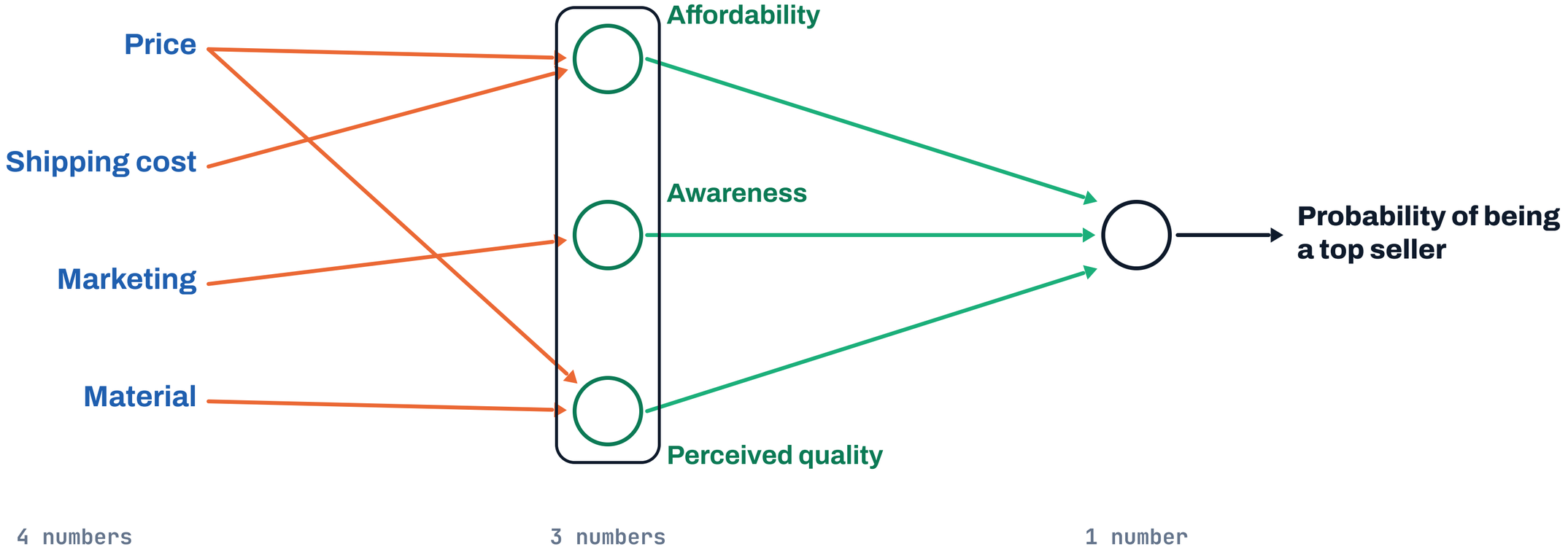


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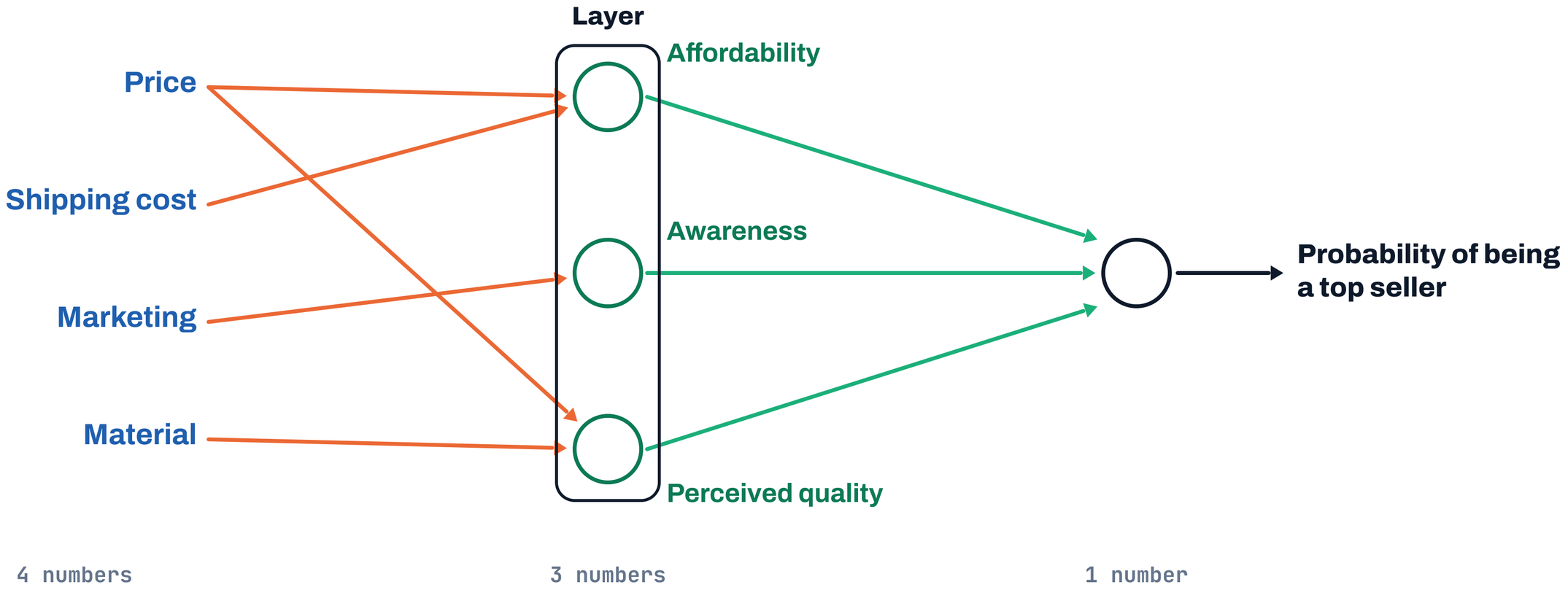


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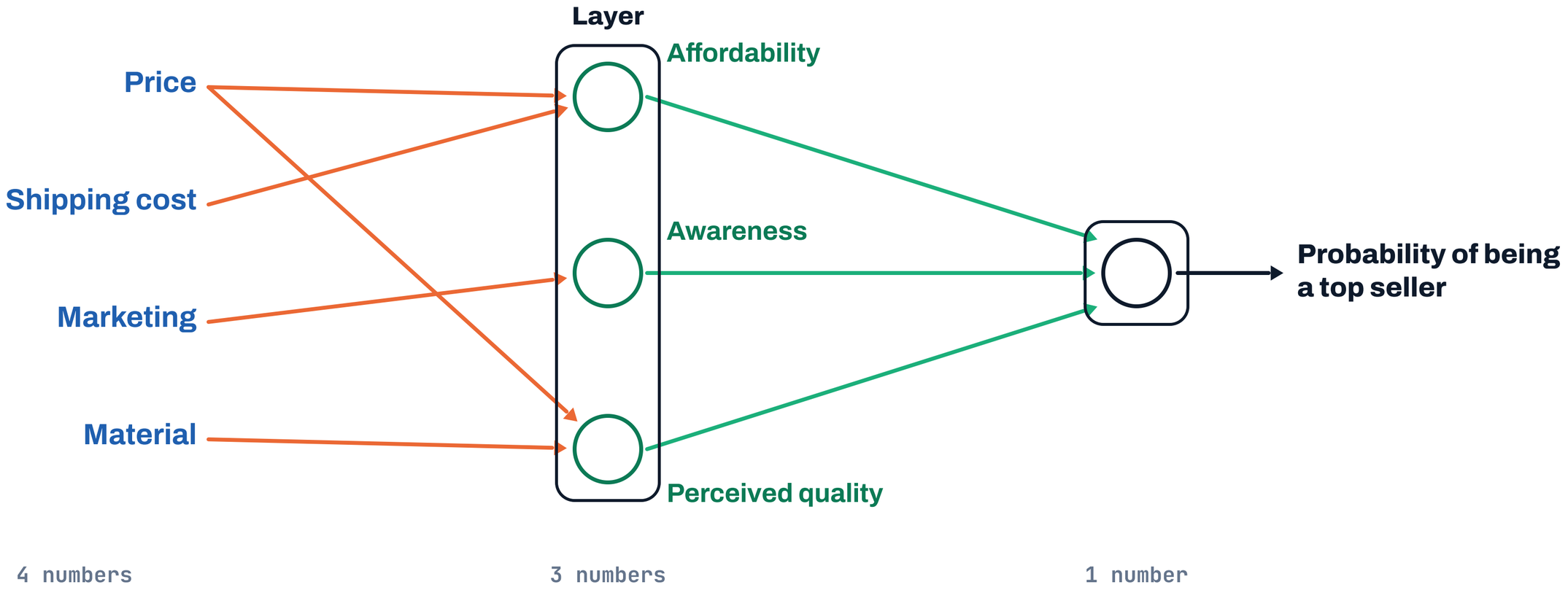


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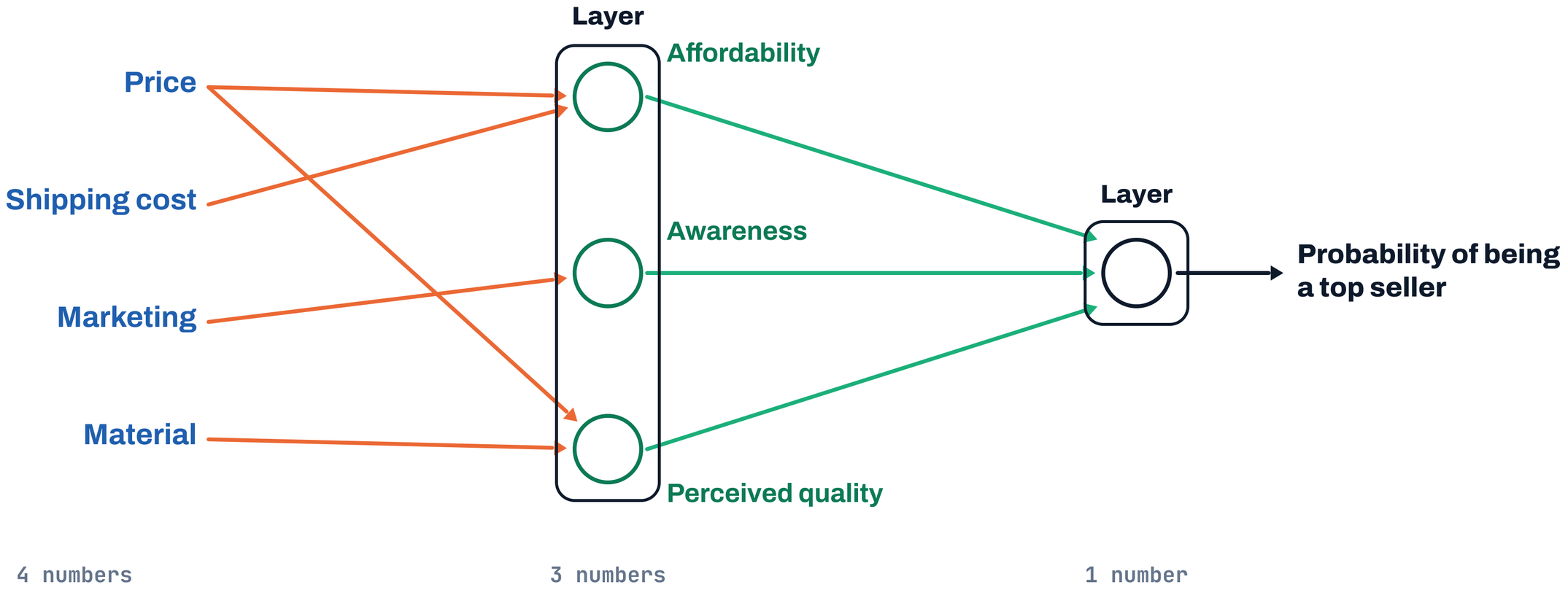


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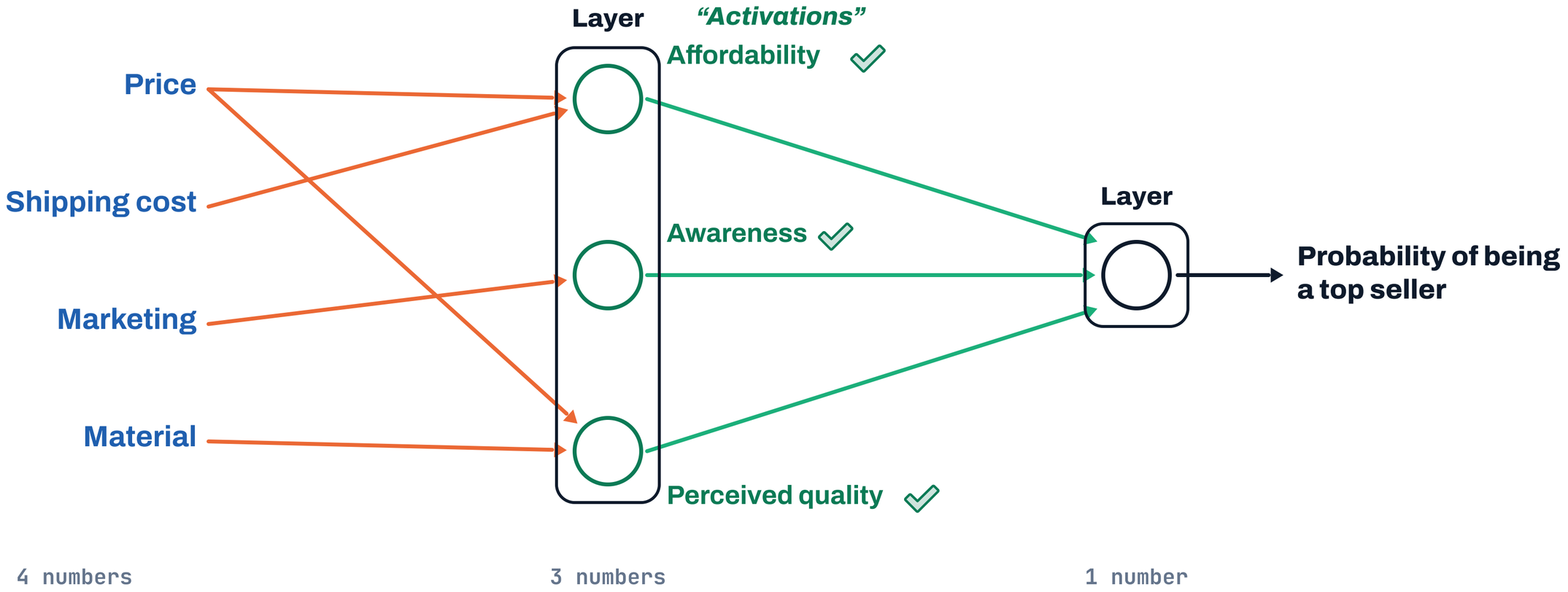


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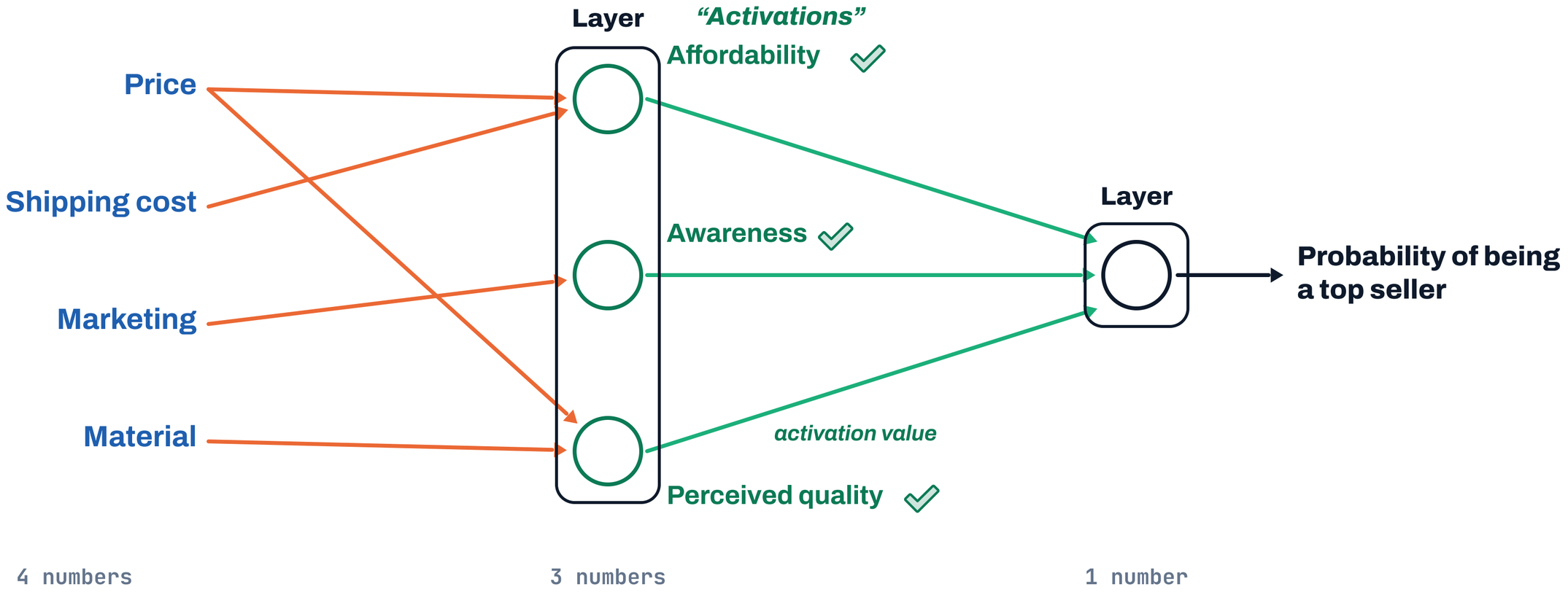


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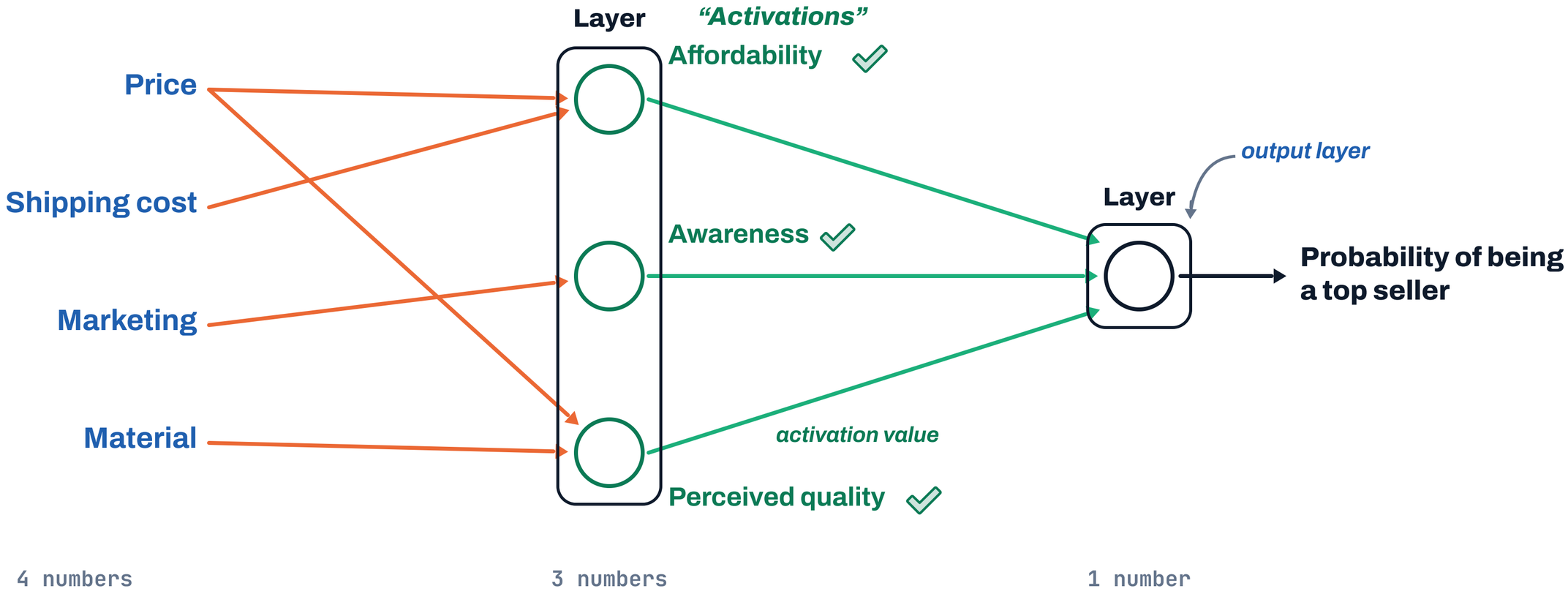


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Layers and activations, one step at a time

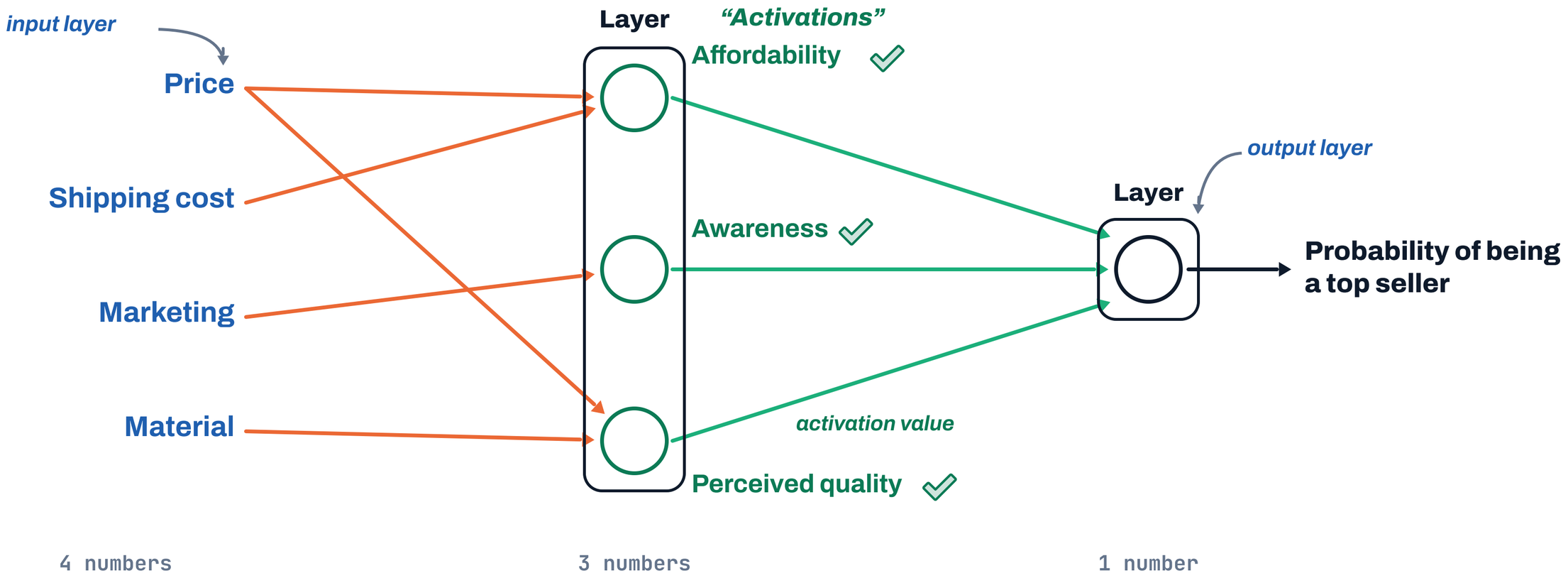


Illustration adapted from Prof. Andrew Ng's lecture (Stanford Online and DeepLearning.AI).

DEMAND PREDICTION · WILL THIS T-SHIRT BE A TOP SELLER?

We chose which inputs feed which neuron

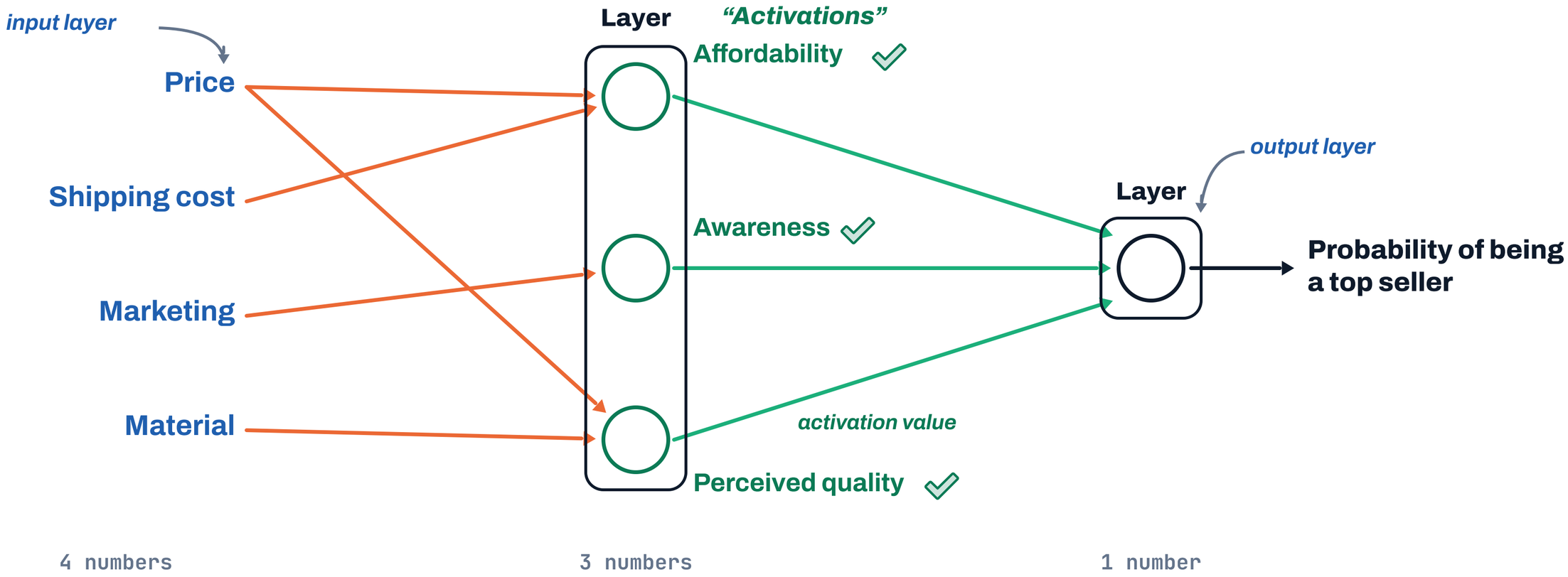


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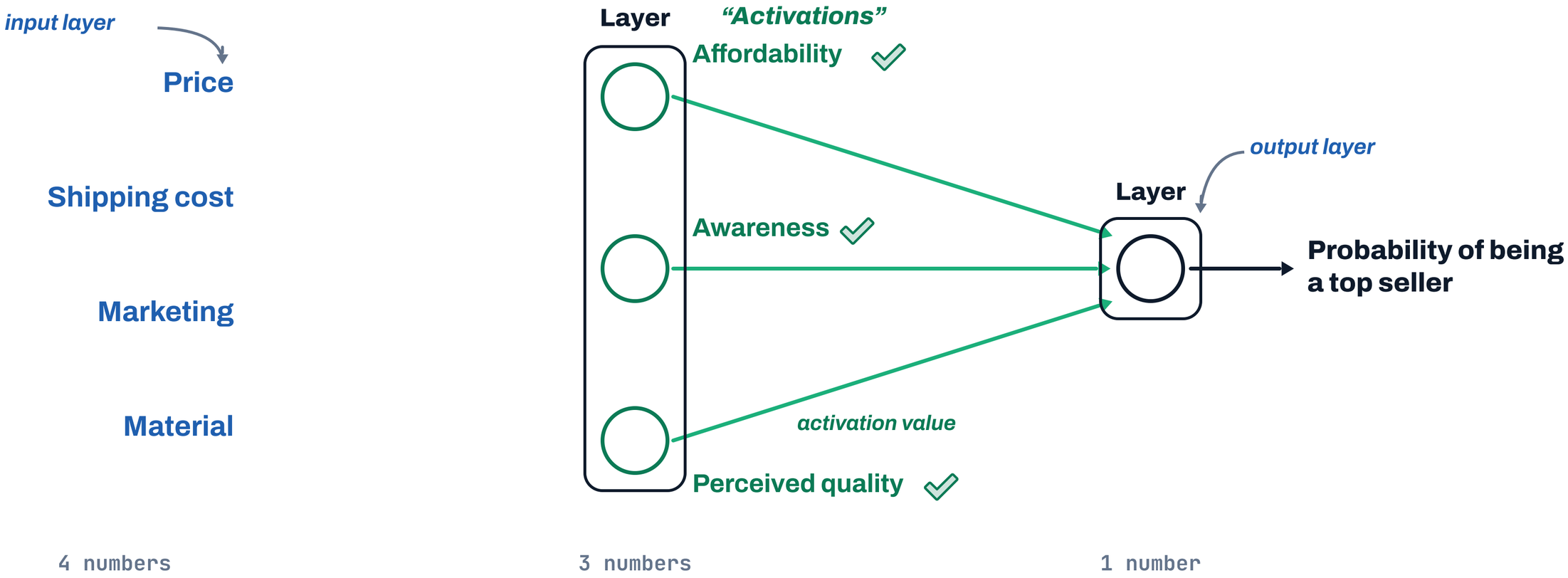


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DEMAND PREDICTION · WILL THIS T-SHIRT BE A TOP SELLER?

In practice, every input feeds every neuron

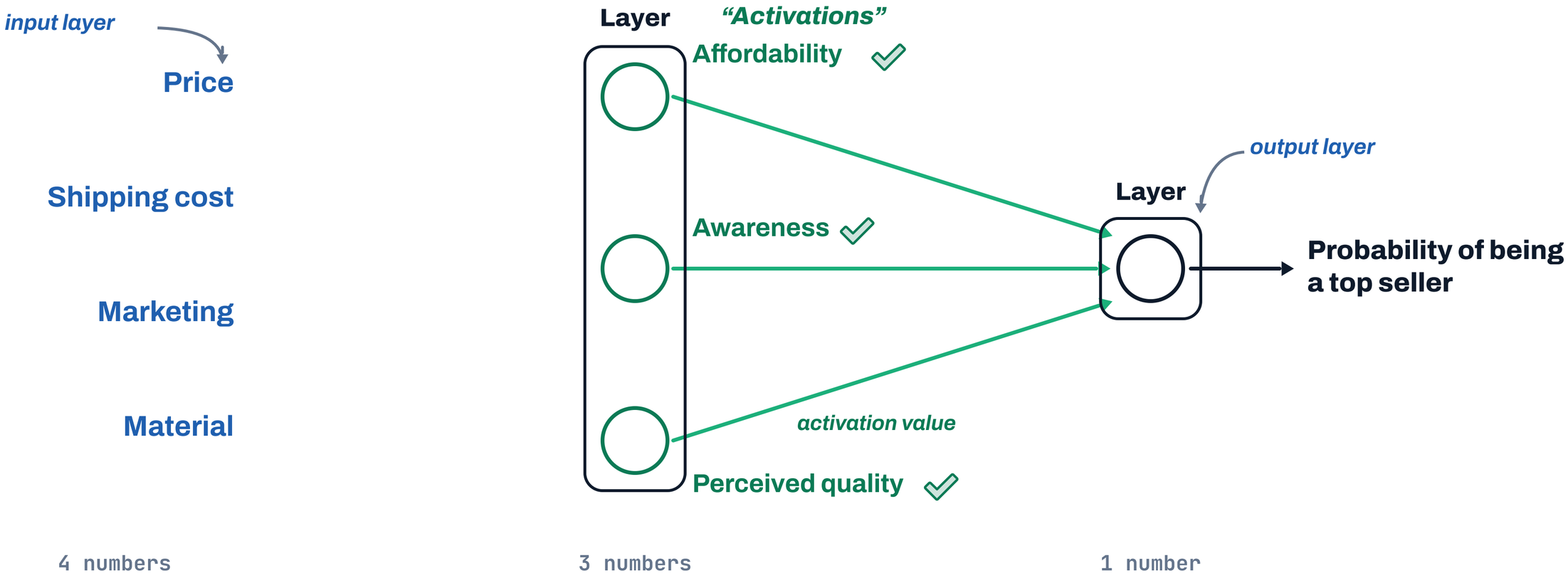


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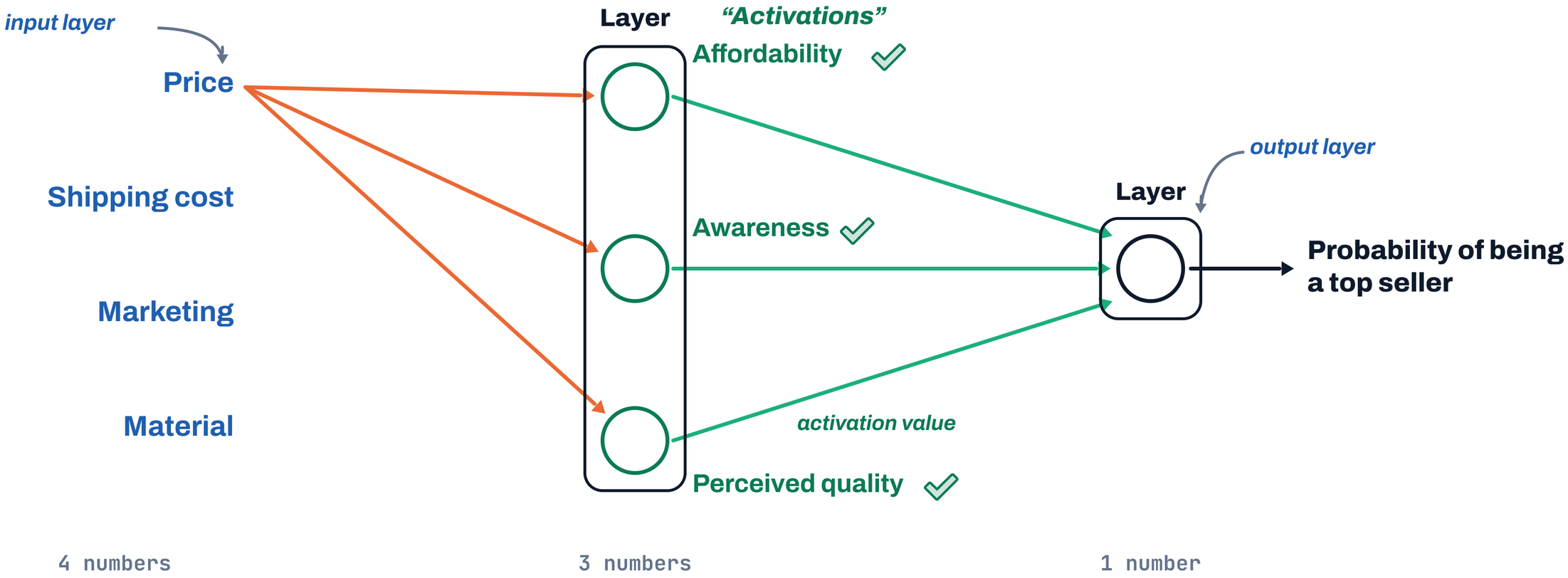


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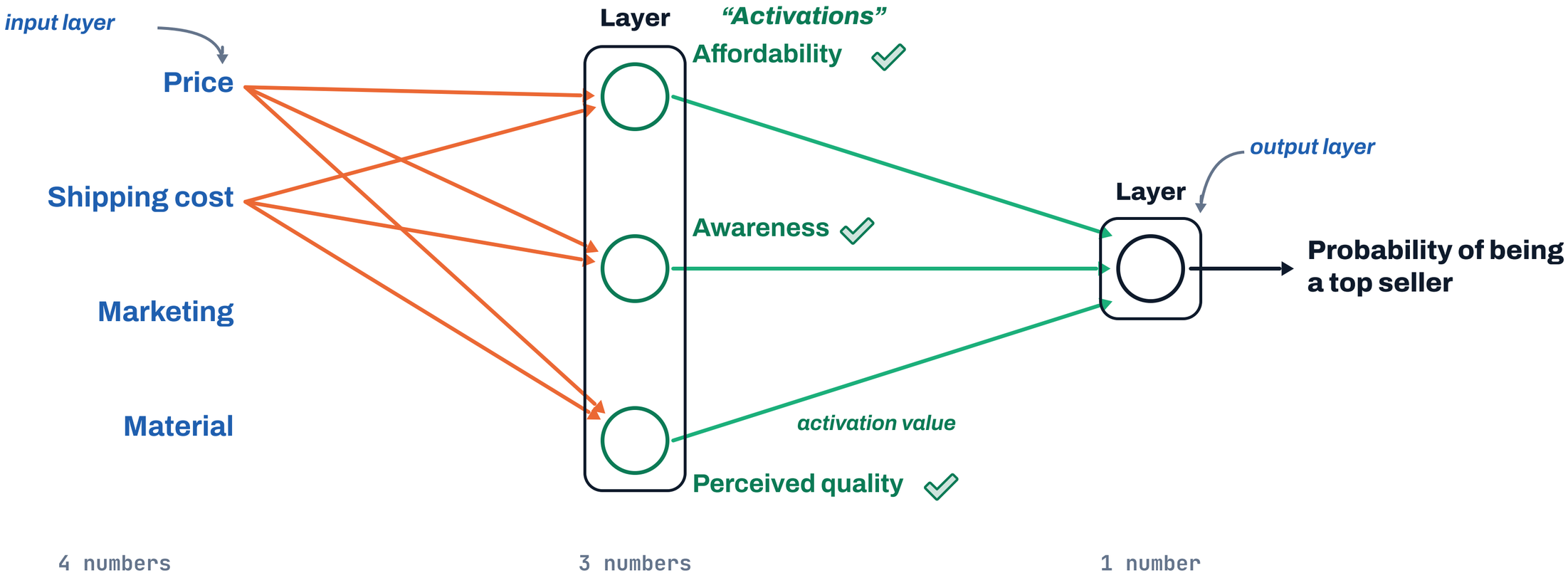


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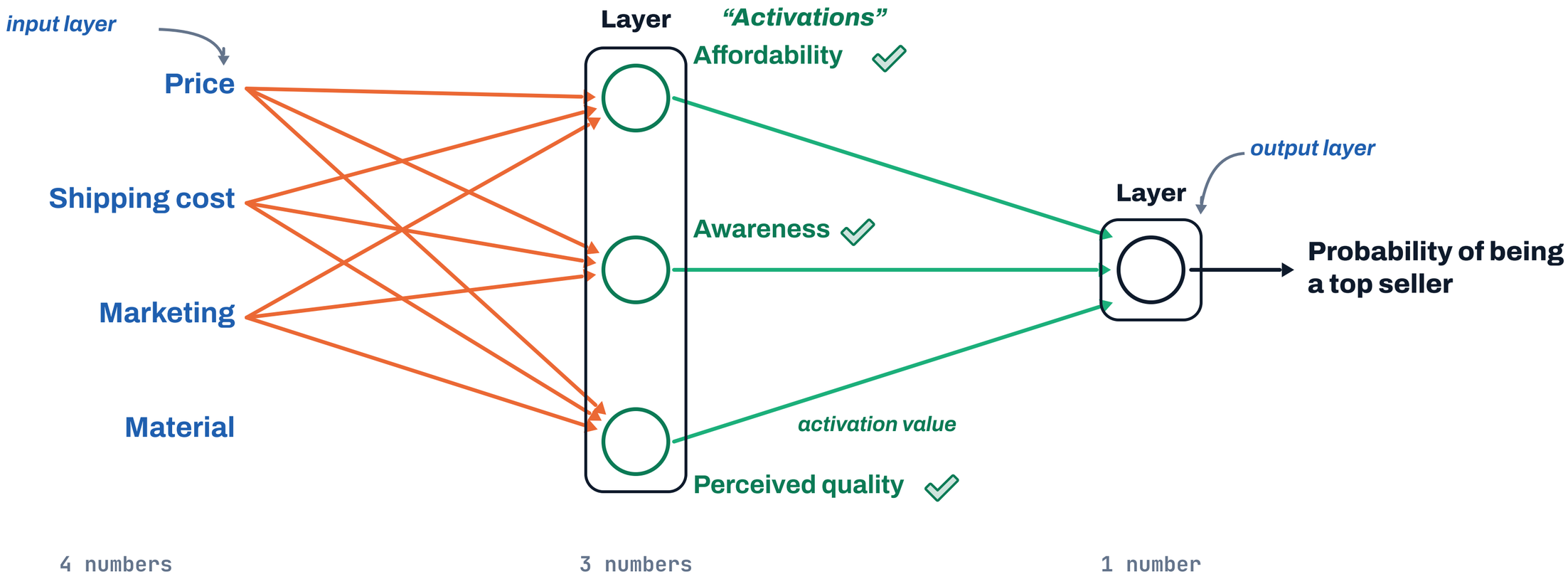


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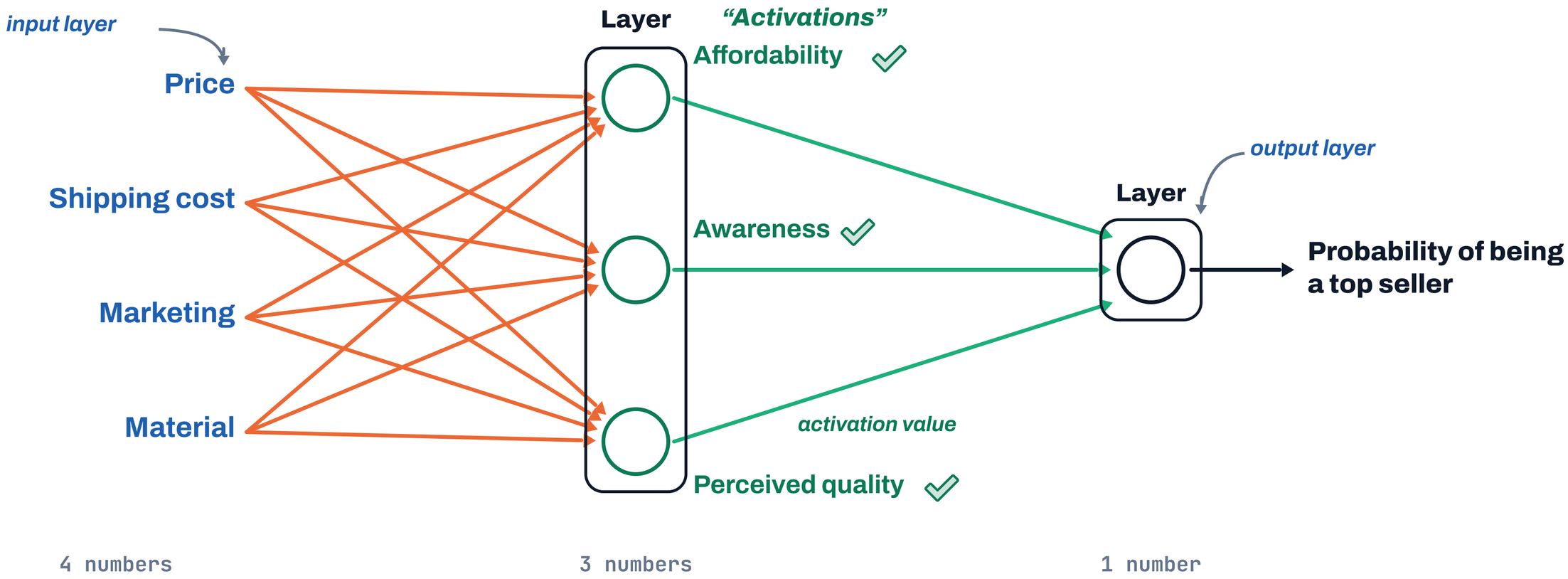


Illustration adapted from Prof. Andrew Ng's lecture (Stanford Online and DeepLearning.AI).

DEMAND PREDICTION · THE SAME NETWORK, ABSTRACTED

A hidden layer turns inputs into new features



Illustration adapted from Prof. Andrew Ng's lecture (Stanford Online and DeepLearning.AI).

DEMAND PREDICTION · THE SAME NETWORK, ABSTRACTED

A hidden layer turns inputs into new features



Instead of you engineering the features by hand, the network **learns its own features** to make prediction easier. This is what makes neural networks **one of the most powerful learning algorithms**.

Illustration adapted from Prof. Andrew Ng's lecture (Stanford Online and DeepLearning.AI).

MULTILAYER PERCEPTRON (MLP)

Architecture: how many layers, how many neurons?

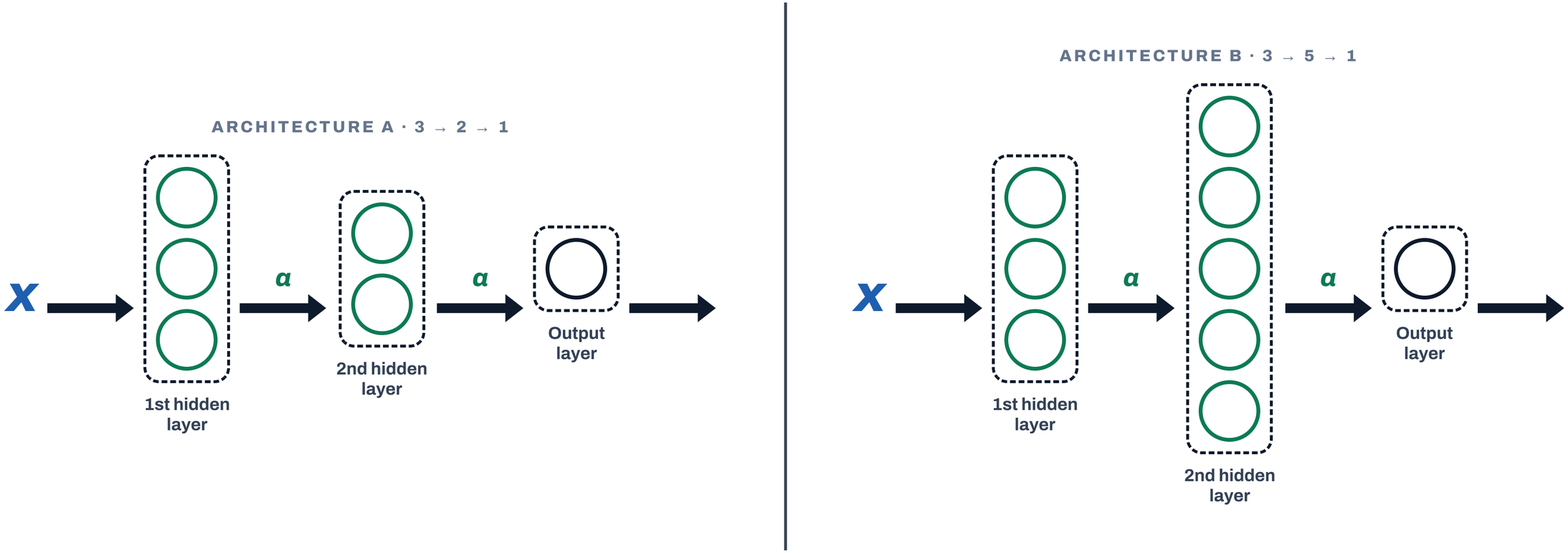
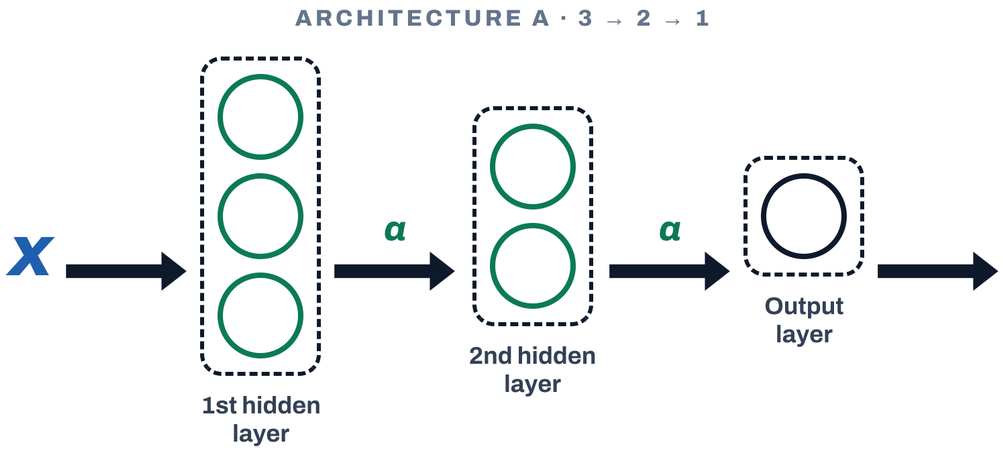


Illustration adapted from Prof. Andrew Ng's lecture (Stanford Online and DeepLearning.AI). Networks of several layers like these are often called multilayer perceptrons.

MULTILAYER PERCEPTRON (MLP)

Architecture: how many layers, how many neurons?

When we build our own network, we decide **how many hidden layers** it has and **how many neurons** each hidden layer has. That choice is the **architecture of the network**.



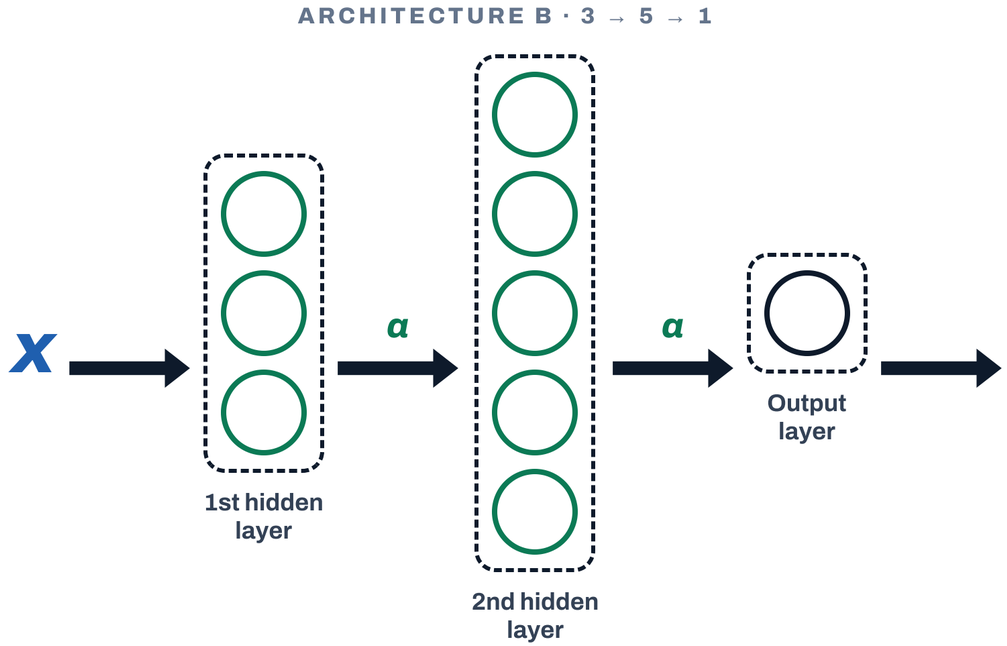
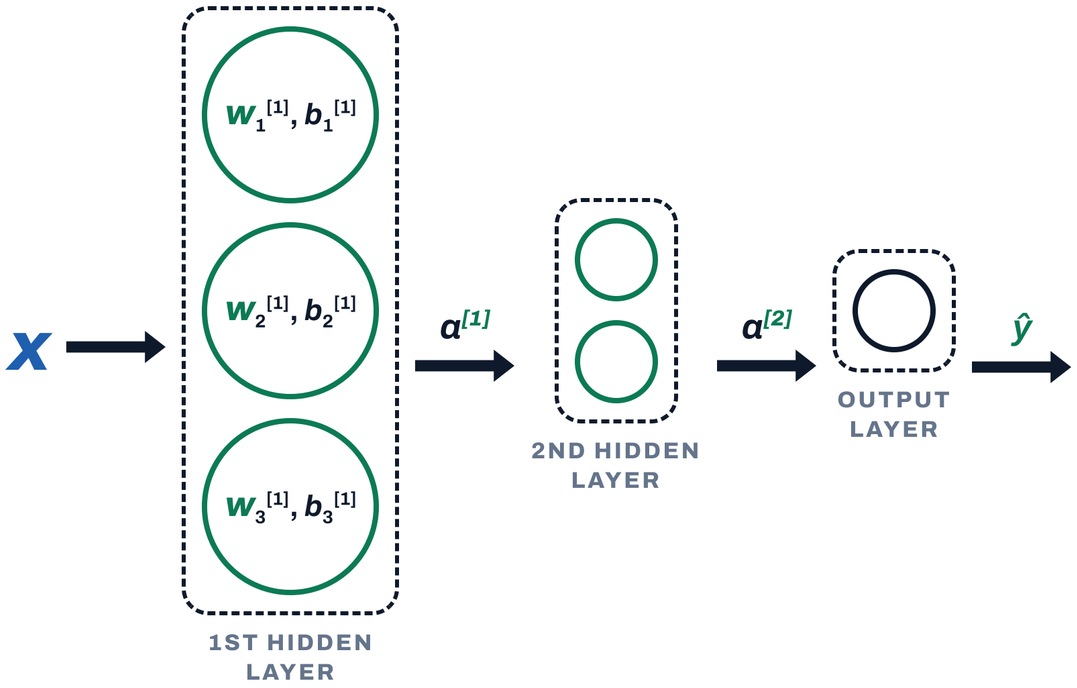


Illustration adapted from Prof. Andrew Ng's lecture (Stanford Online and DeepLearning.AI). Networks of several layers like these are often called multilayer perceptrons.

MULTILAYER PERCEPTRON (MLP) · INSIDE A LAYER

Every neuron has its own weights and bias



LAYER 1, NEURON BY NEURON

1 $a_1^{[1]} = g(\mathbf{w}_1^{[1]} \cdot \mathbf{x} + b_1^{[1]})$

2 $a_2^{[1]} = g(\mathbf{w}_2^{[1]} \cdot \mathbf{x} + b_2^{[1]})$

3 $a_3^{[1]} = g(\mathbf{w}_3^{[1]} \cdot \mathbf{x} + b_3^{[1]})$

$\mathbf{a}^{[1]} = (a_1, a_2, a_3)$ is the input of layer 2. Superscript $[l]$ = layer, subscript j = neuron, g = activation function, bold = vector.

Illustration adapted from Prof. Andrew Ng's lecture (Stanford Online and DeepLearning.AI).

KEY TRAINING TERMS IN AN ANN (MLP) · 1 OF 3

How long do we train?

KEY TRAINING TERMS IN AN ANN (MLP) · 1 OF 3

How long do we train?

1

Epoch



One epoch: **every training row is used once.**

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One epoch: **every training row is used once.**

DEFINITION

An epoch is **one complete pass** through the entire training dataset by the model.

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ANALOGY

Like reading the **whole book once**.

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One epoch: **every training row is used once.**

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An epoch is **one complete pass** through the entire training dataset by the model.

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Like reading the **whole book once**.

IN PRACTICE

We train for **many epochs** (10, 50, 100, ...) so the model keeps improving.

Within each epoch the weights are updated again and again, once per batch.

KEY TRAINING TERMS IN AN ANN (MLP) · 2 OF 3

How much data per update?

KEY TRAINING TERMS IN AN ANN (MLP) · 2 OF 3

How much data per update?

2

Batch size



Example: **800 rows, batch size 32 → 25 weight updates** per epoch.

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Example: **800 rows, batch size 32 → 25 weight updates** per epoch.

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The **number of training samples** the model processes **before updating the weights once**.

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Example: **800 rows, batch size 32 → 25 weight updates** per epoch.

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The **number of training samples** the model processes **before updating the weights once**.

WHY IT MATTERS

The model does not always look at the whole dataset at once.

It breaks the data into **smaller batches**. Common values: **32, 64, 128**.

KEY TRAINING TERMS IN AN ANN (MLP) · 2 OF 3

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It breaks the data into **smaller batches**. Common values: **32, 64, 128**.

IN PRACTICE

Small batch: more updates, noisier learning, but faster to start.

Large batch: smoother updates, more memory.

KEY TRAINING TERMS IN AN ANN (MLP) · 3 OF 3

How big is each step?

KEY TRAINING TERMS IN AN ANN (MLP) · 3 OF 3

How big is each step?

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Learning rate

Each update moves the weights a step **downhill on the loss**. The learning rate sets the step size.

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A small number that controls **how big a step** the model takes while updating the weights during training.

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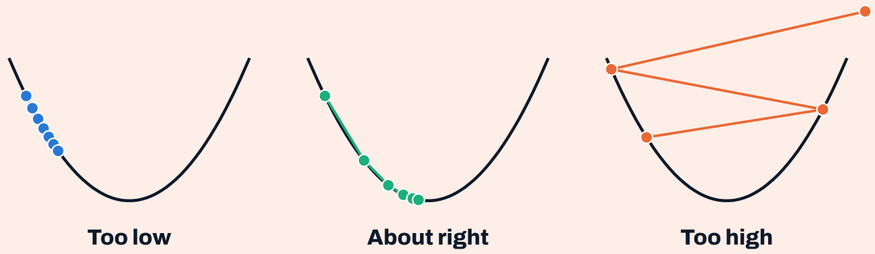
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Too high: the model may overshoot the optimum and not converge.

Too low: the model learns very slowly and may get stuck.



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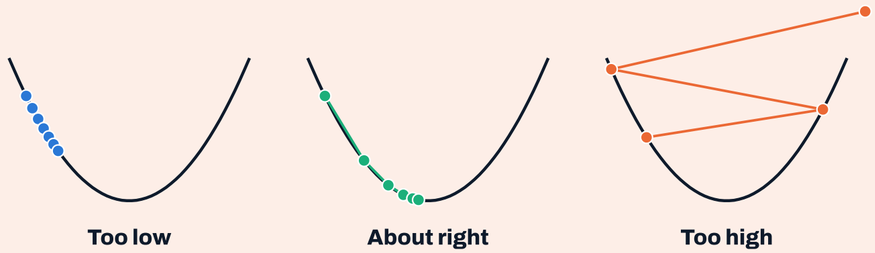
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ANALOGY

Like your **pace while hiking** towards a goal: not too fast, not too slow.

KEY TRAINING TERMS · SUMMARY

Three settings that shape training

TERM	WHAT IT MEANS	ANALOGY	IN SCIKIT-LEARN'S MLPREGRESSOR
Epoch	One full pass through the training data	Reading the whole book once	`max_iter` maximum number of epochs (adam, sgd)
Batch size	Samples used in one weight update	Reading one chapter at a time	`batch_size="auto"` min(200, number of rows)
Learning rate	Step size for updating the weights	How big a step you take	`learning_rate_init=0.001` the default

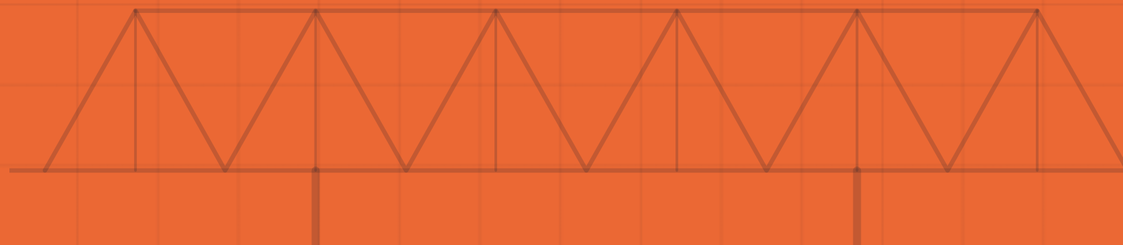
These are hyperparameters: we choose them, the network does not learn them. Choose them on validation data.

02

- a What an ANN is, and one neuron
- b Reading the (32, 16) architecture
- c Why nonlinear activations matter
- d Scaling the inputs
- e How the network learns
- f Data roles and choosing the architecture

PART 02 · BUILD & TRAIN

An ANN for concrete strength



WHAT IS AN ARTIFICIAL NEURAL NETWORK?

A parameterised function built from connected units

Inputs

The measured features enter the network.

`8 mix and age variables`

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32 and 16 neurons

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`ŷ in MPa`

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The measured features enter the network.

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`32 and 16 neurons`

Output

One predicted strength value.

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The network learns its weights and biases from examples. It is not handed an explicit engineering equation.

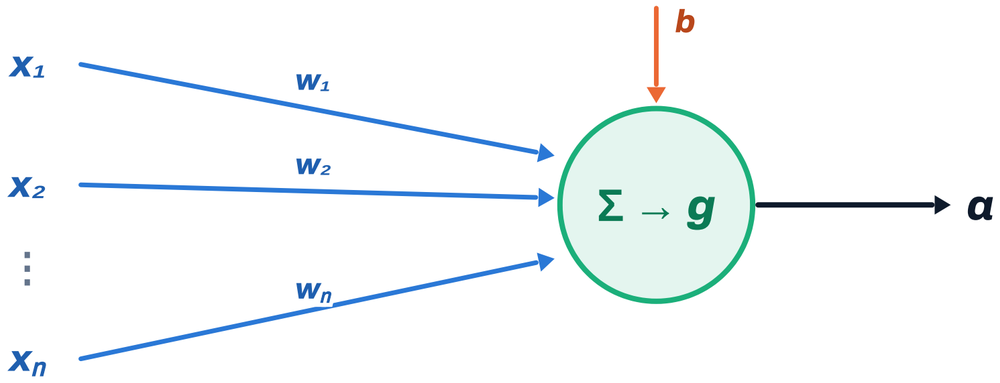
INSIDE ONE NEURON

Combine the inputs, add a bias, apply an activation

TWO STEPS

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

$$a = g(z)$$



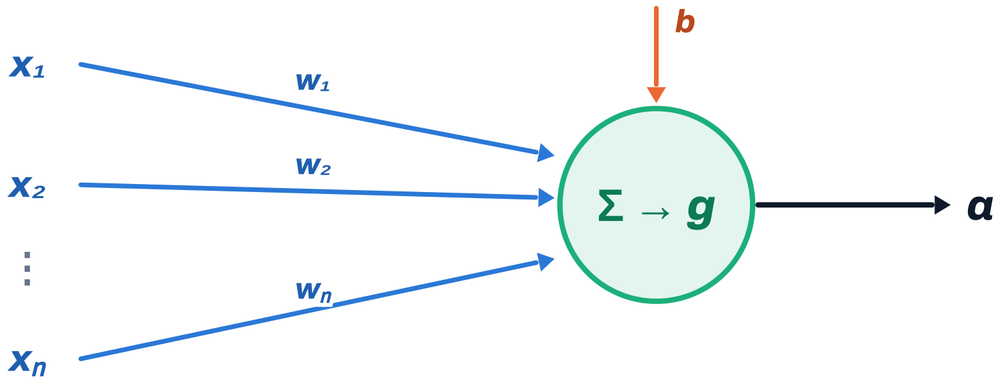
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WEIGHTS W

Control how much each input contributes.

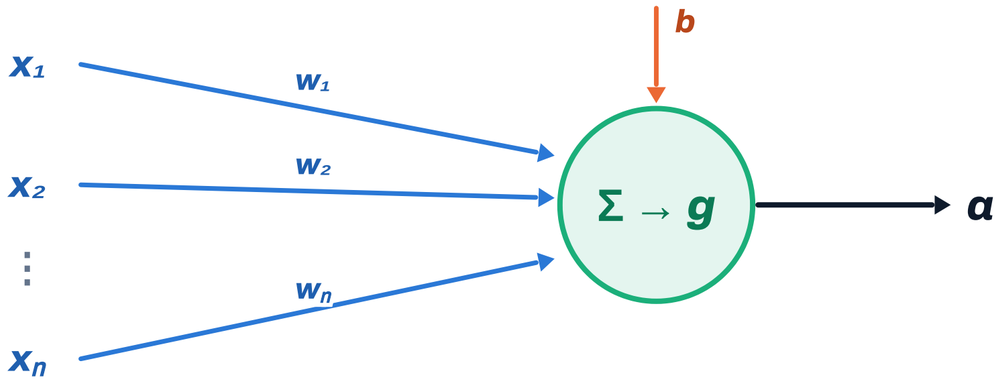
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BIAS B

Shifts the pre-activation value z.

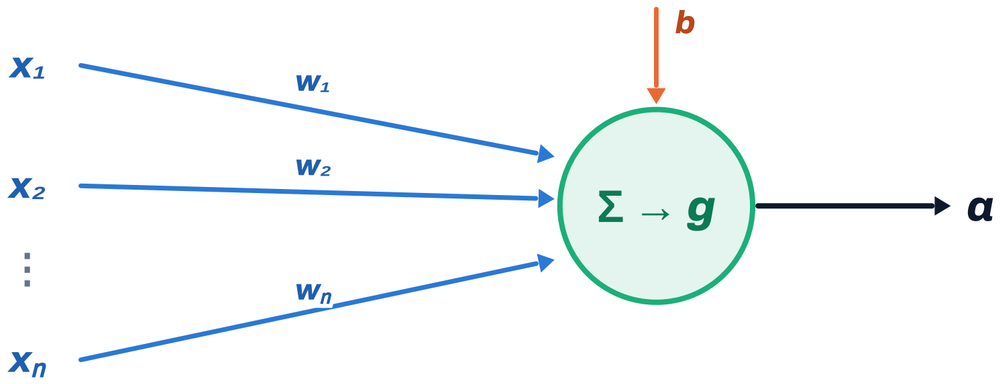
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ACTIVATION G

Bends the response, so layers can model curves.

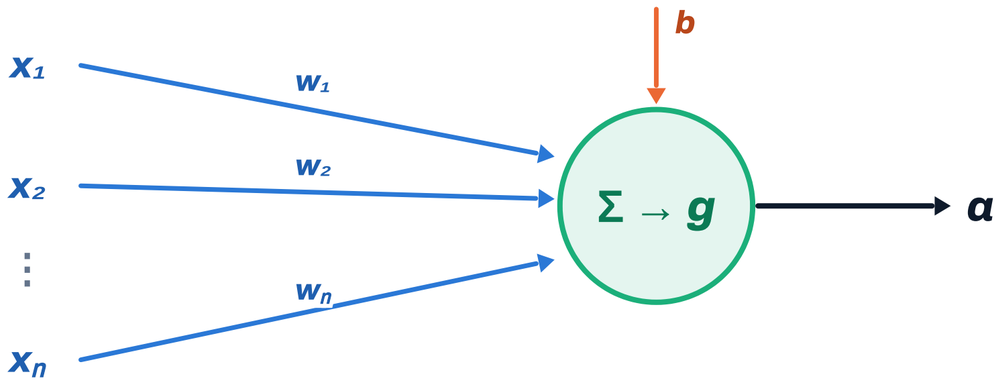
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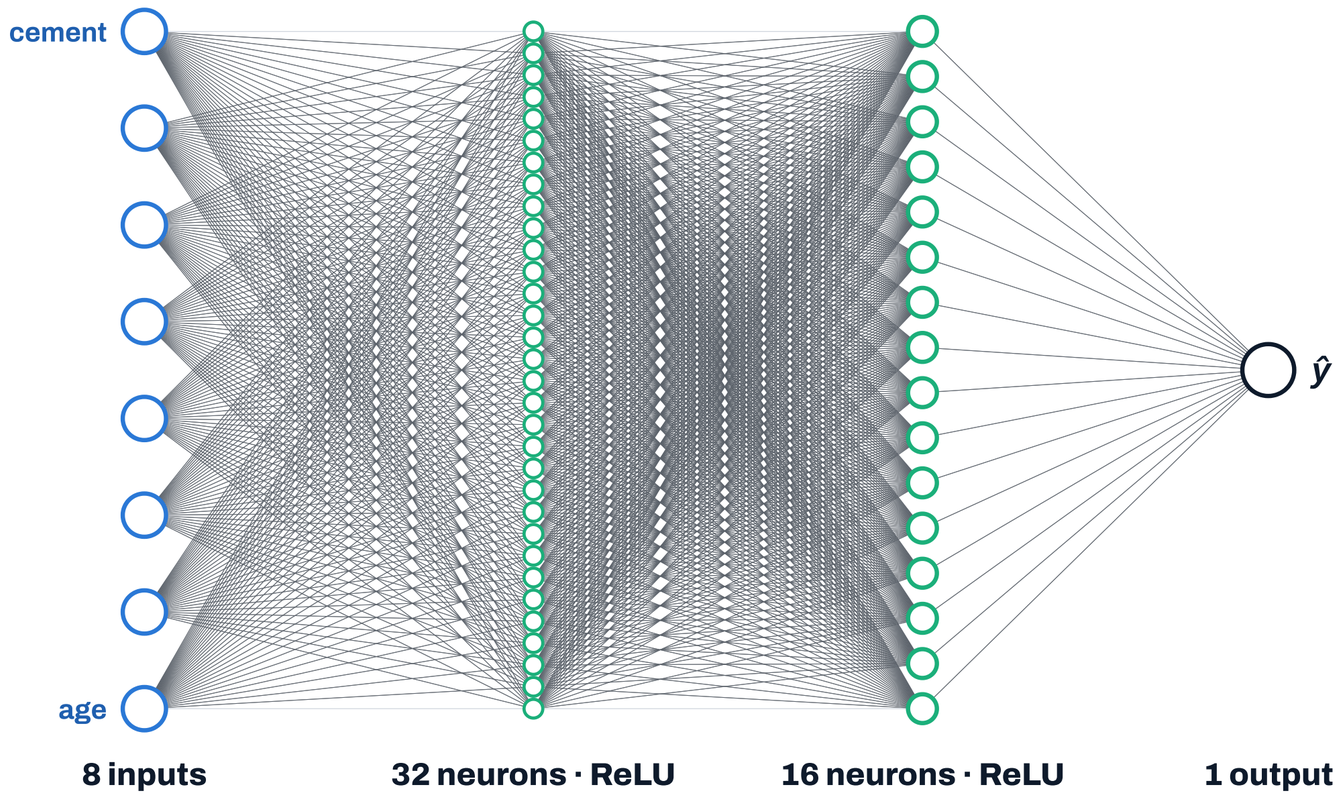
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Shifts the pre-activation value z.

ACTIVATION G
Bends the response, so layers can model curves.

Training adjusts every weight and every bias. The same z appeared in Session 05 as the logistic-regression score.

READING AN ANN ARCHITECTURE

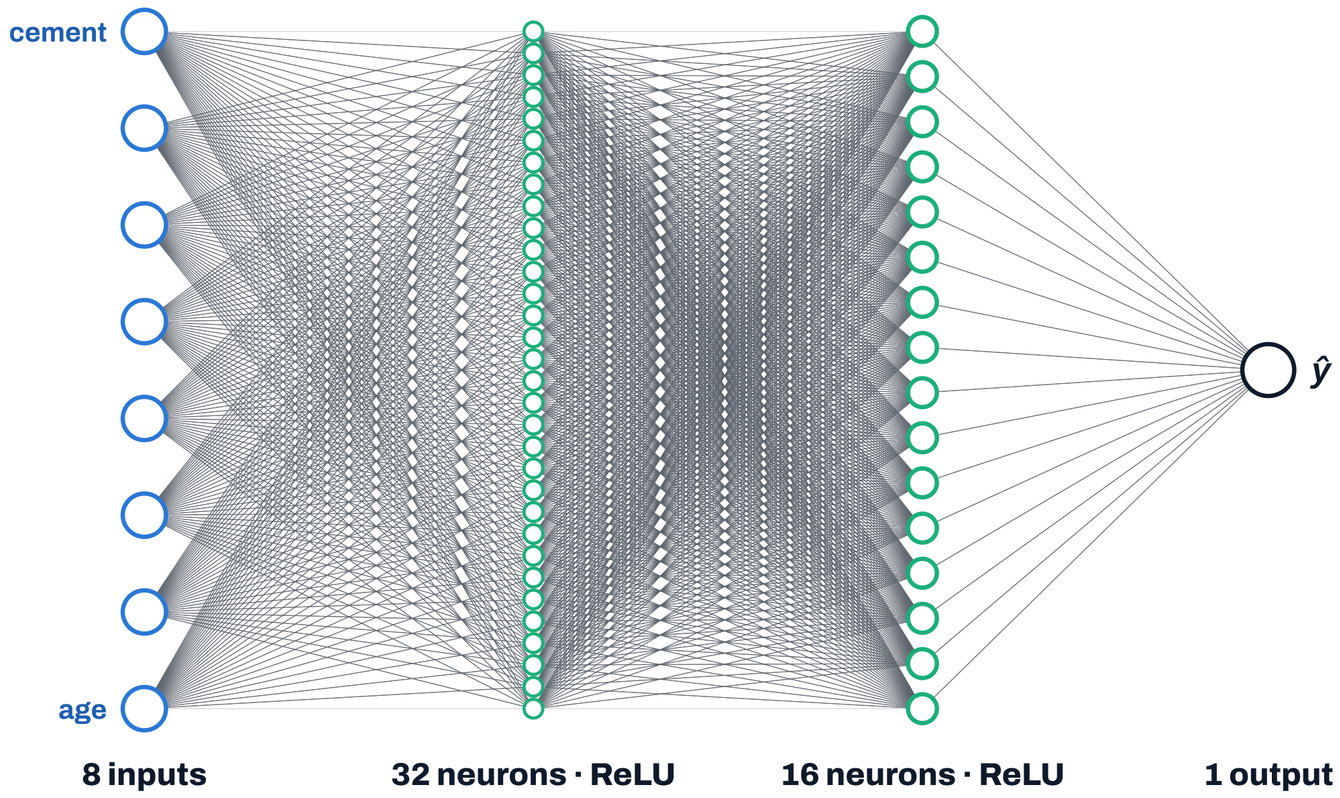
Our selected network is (32, 16)



WEIGHTS AND BIASES TO LEARN

READING AN ANN ARCHITECTURE

Our selected network is (32, 16)

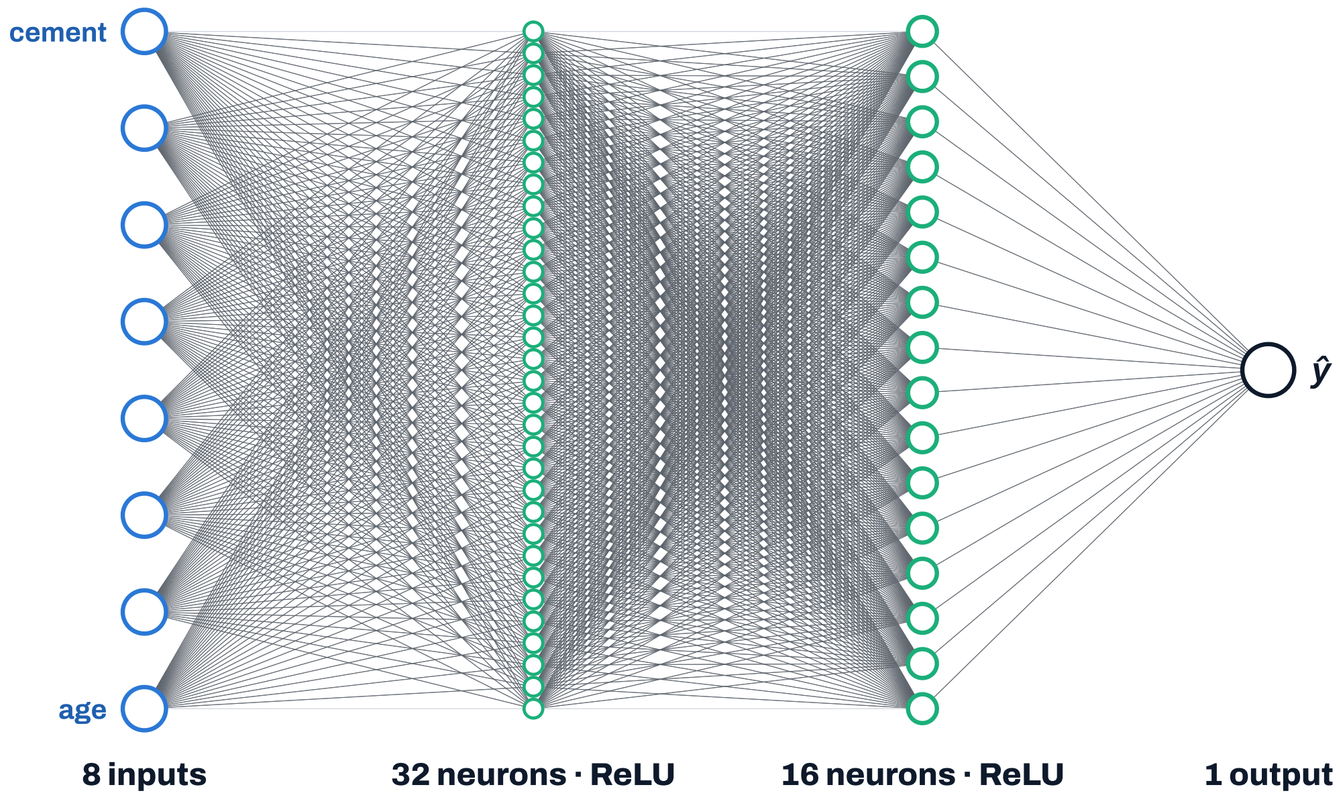


WEIGHTS AND BIASES TO LEARN

Inputs → 32 · 8 × 32 + 32 **288**

READING AN ANN ARCHITECTURE

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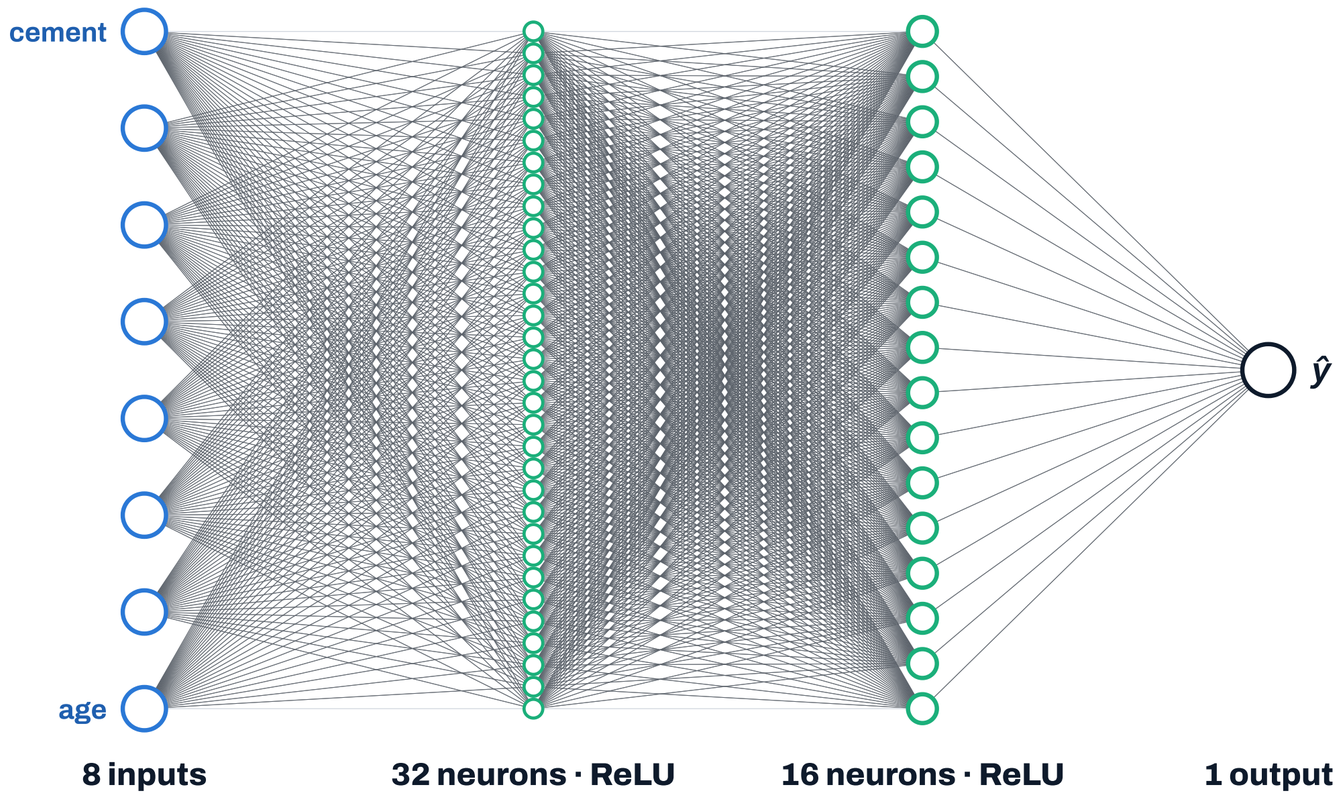


WEIGHTS AND BIASES TO LEARN

Inputs → 32 · $8 \times 32 + 32$	**288**
32 → 16 · $32 \times 16 + 16$	**528**

READING AN ANN ARCHITECTURE

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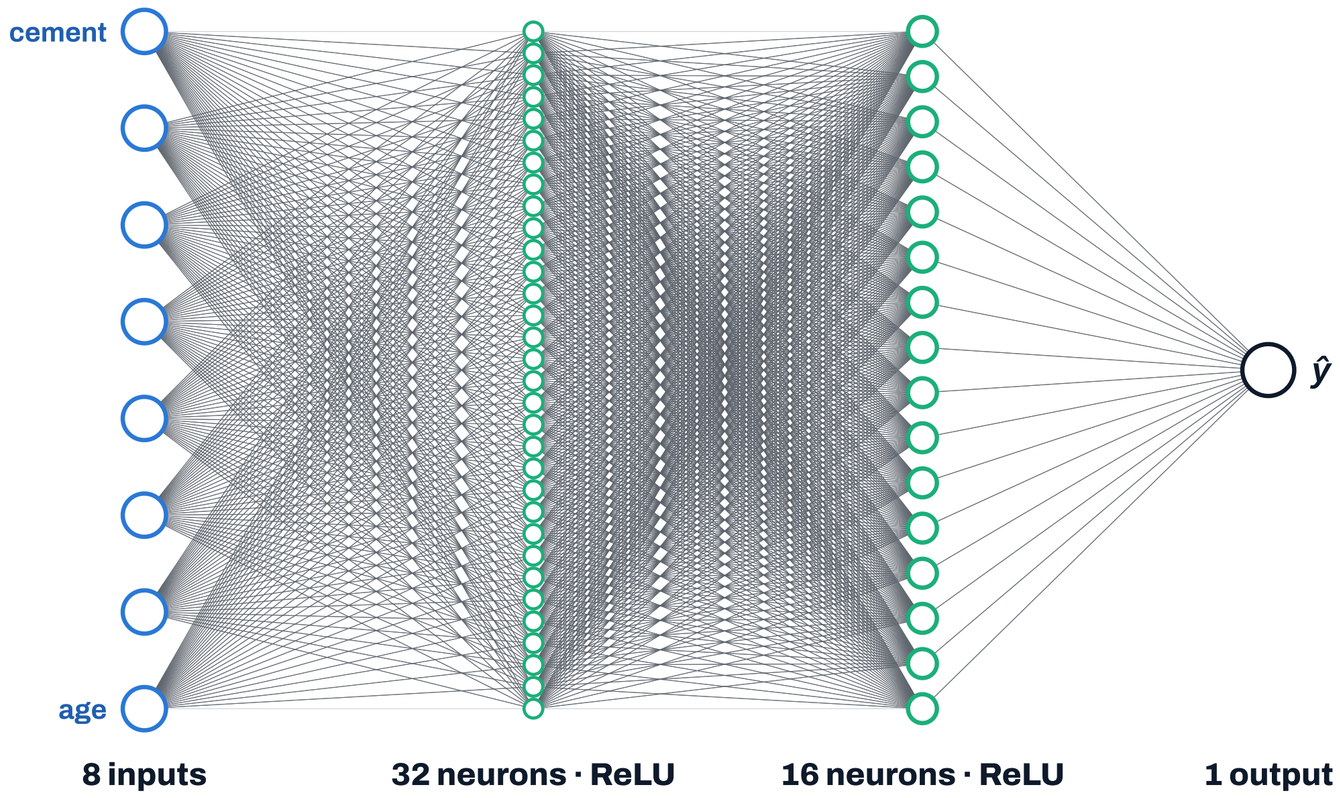


WEIGHTS AND BIASES TO LEARN

Inputs → 32 · 8 × 32 + 32	**288**
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16 → output · 16 × 1 + 1	**17**

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WEIGHTS AND BIASES TO LEARN

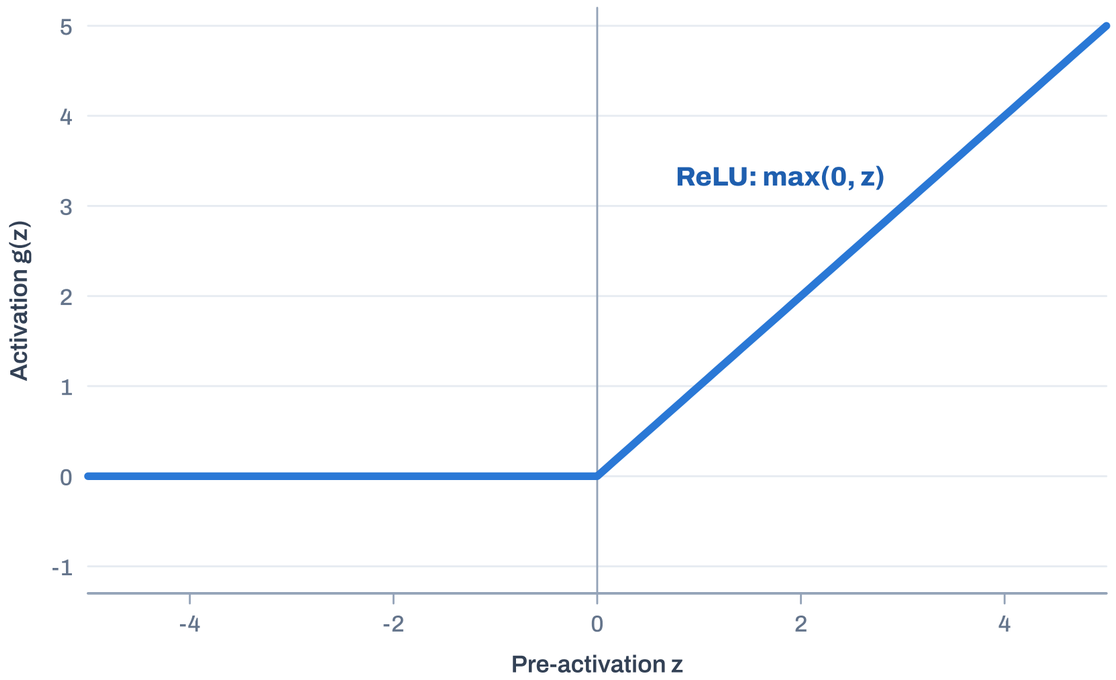
Inputs → 32 · 8 × 32 + 32	288
32 → 16 · 32 × 16 + 16	528
16 → output · 16 × 1 + 1	17
Total parameters	833

(32, 16) gives the neurons in each hidden layer, not the number of parameters. Two hidden layers here.

WHY NONLINEAR ACTIVATION FUNCTIONS MATTER

Without them, stacked layers stay linear

Three common activation functions



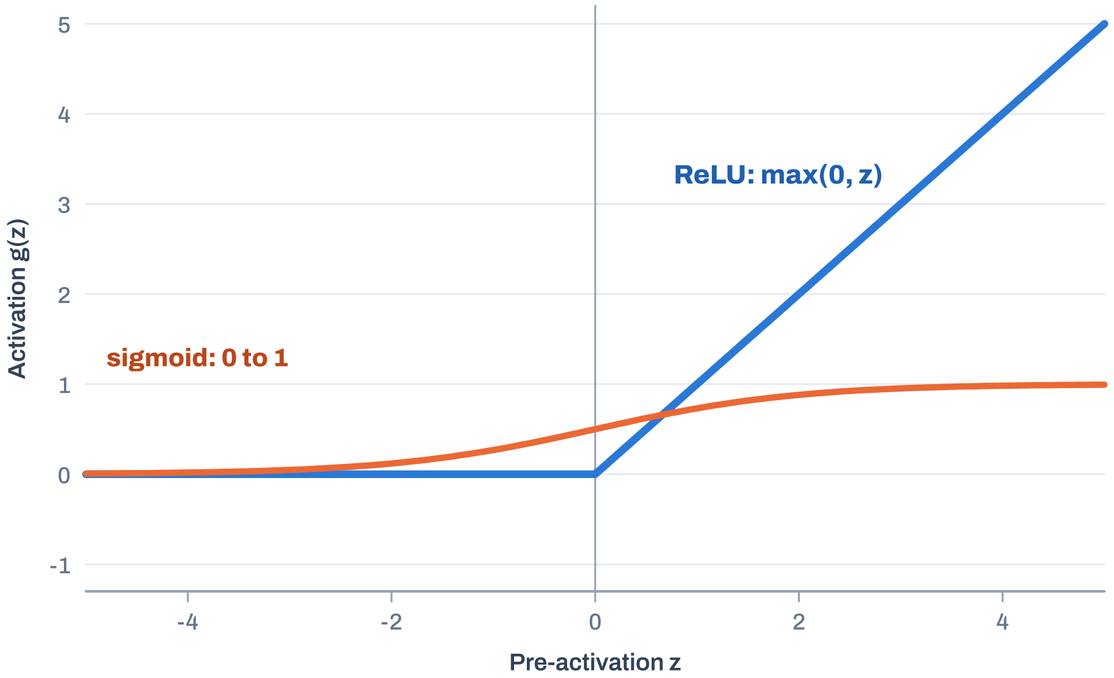
IN OUR FITTED ANN

Both hidden layers use **ReLU**. The output neuron is linear, so ŷ can be any strength.

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WHY NOT LINEAR EVERYWHERE?

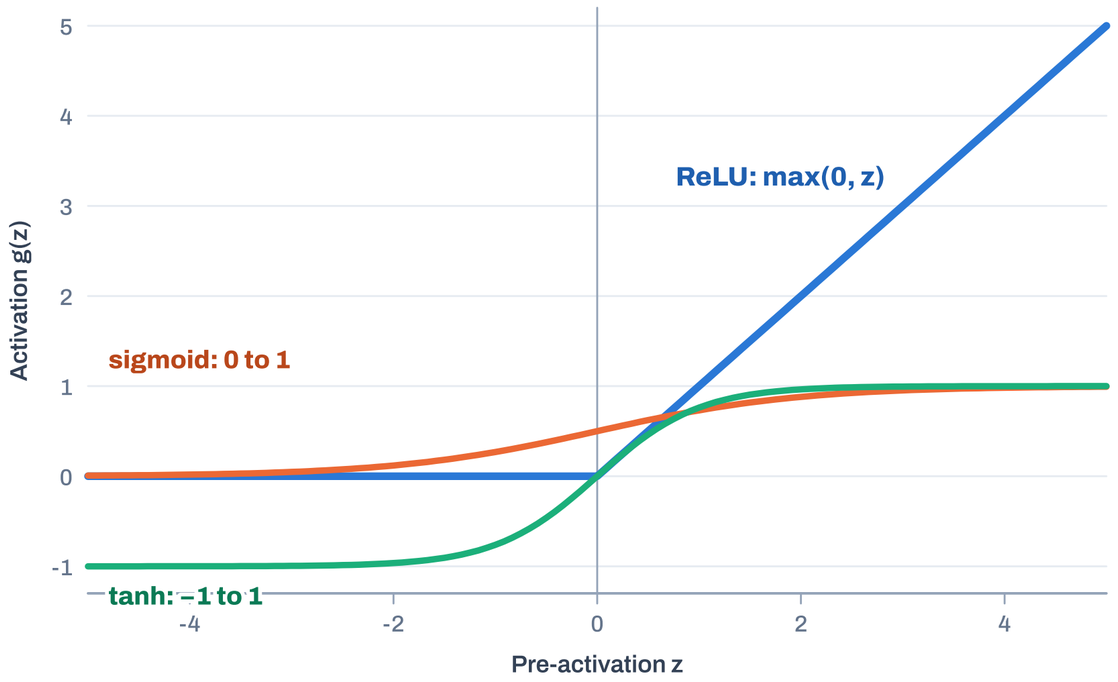
$$W_2(W_1x + b_1) + b_2 = Wx + b$$

Two linear layers collapse into one linear model.

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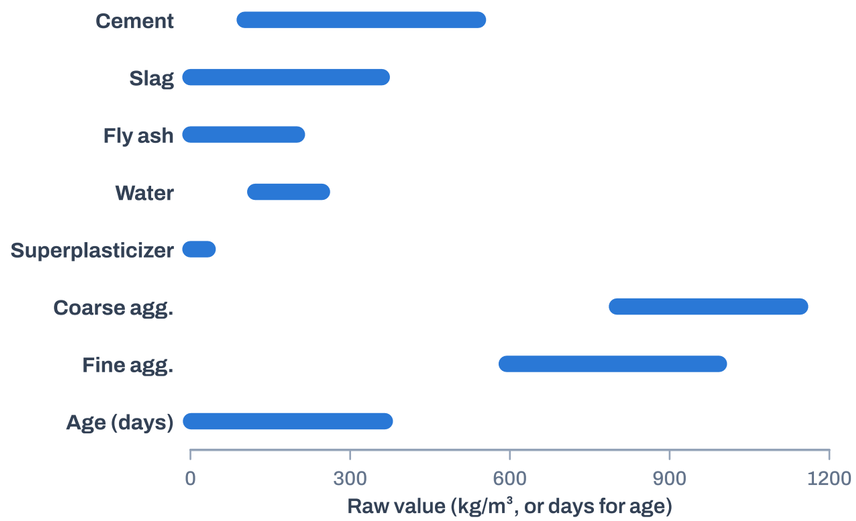
Sigmoid and tanh are shown for comparison; they are not used in this fitted model.

WHY SCALE THE INPUTS?

Our features use very different ranges and units

Raw inputs

Superplasticizer spans 0–32.2 kg/m³; coarse aggregate 801–1,145 kg/m³.

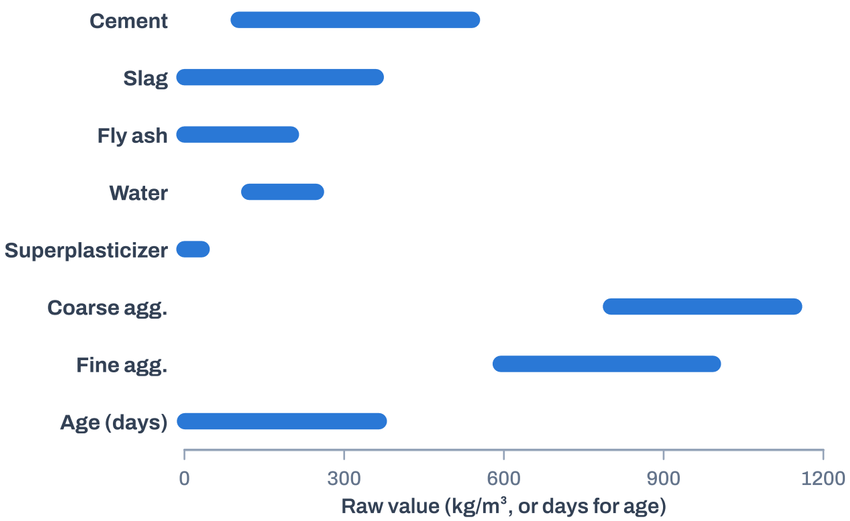


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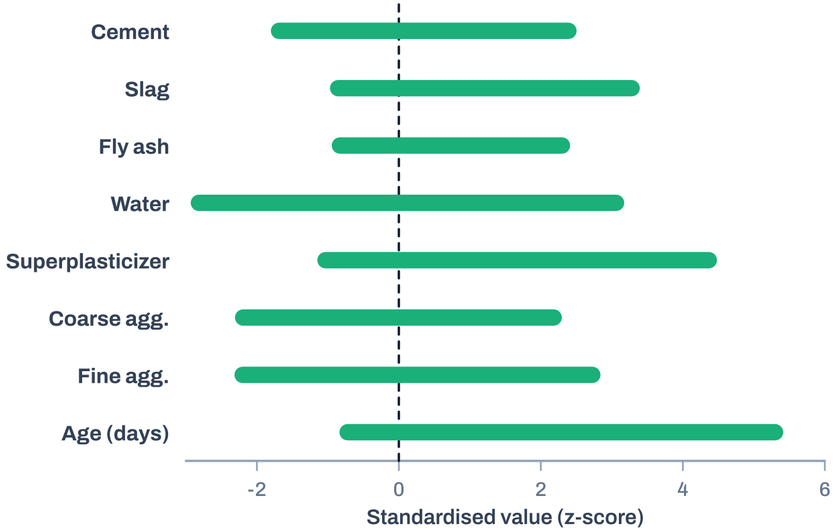
Raw inputs

Superplasticizer spans 0–32.2 kg/m³; coarse aggregate 801–1,145 kg/m³.



After StandardScaler

Each feature centred and scaled with the training mean and standard deviation.

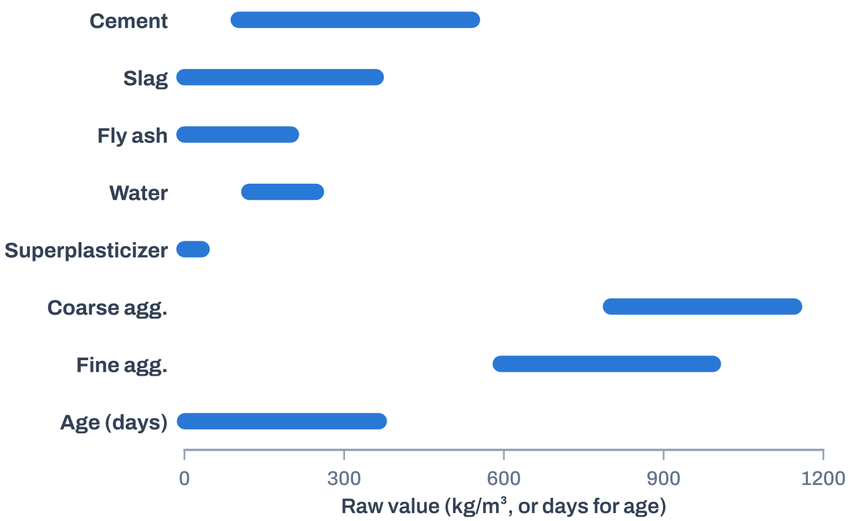


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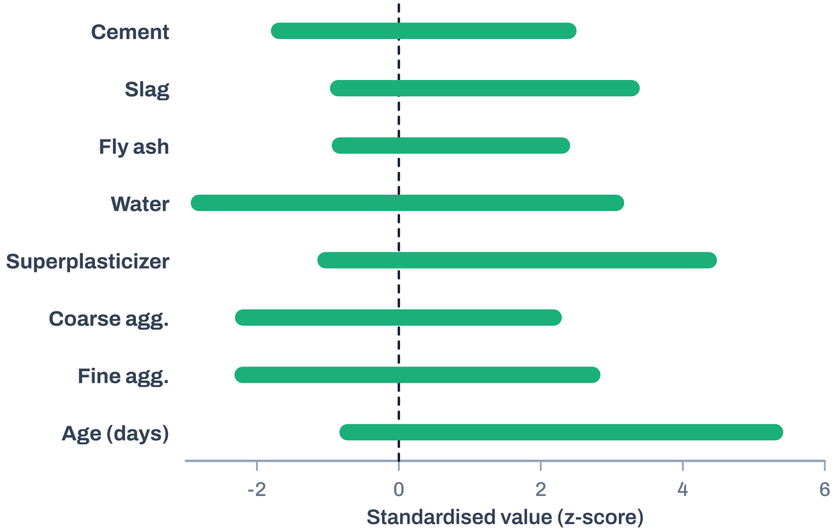
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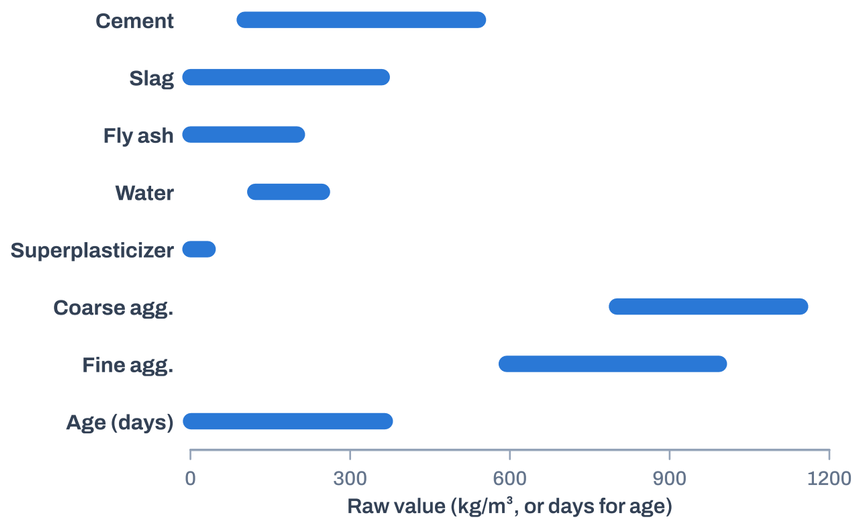
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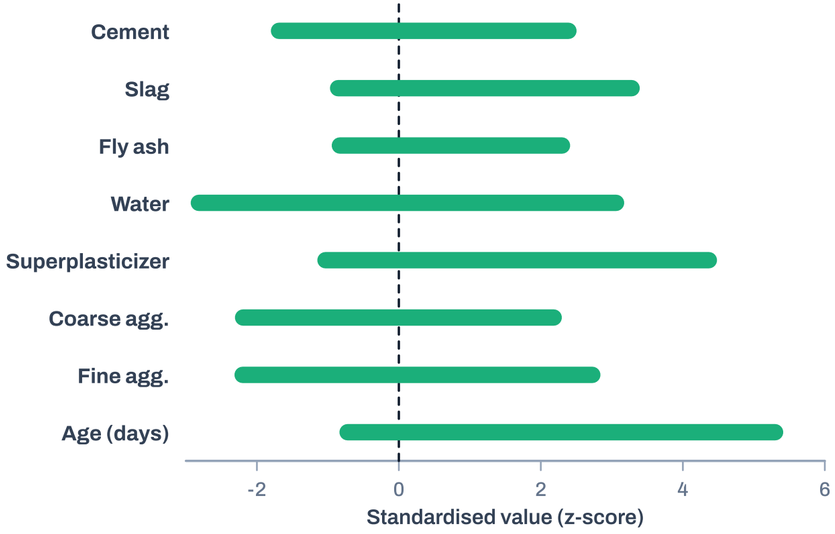
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The scaler sits inside a Pipeline, so it is fitted on the training data, never on the held-out test set.

HOW DOES AN ANN LEARN?

Adjust weights and biases to reduce a loss

1

Forward pass

The inputs flow through the layers and produce a prediction ŷ.

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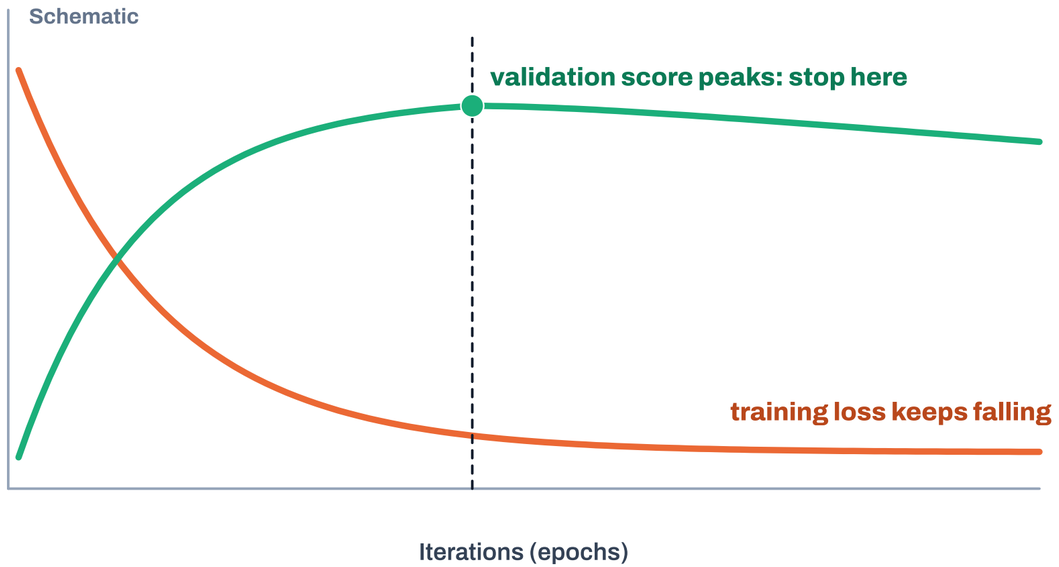
This is optimisation, the same gradient descent as Session 03. It is not the discovery of a guaranteed physical law.

TRAINING LOSS AND VALIDATION SCORE

They answer different questions

What the notebook’s training plot shows

Early stopping keeps the iteration with the best validation score.



TRAINING LOSS

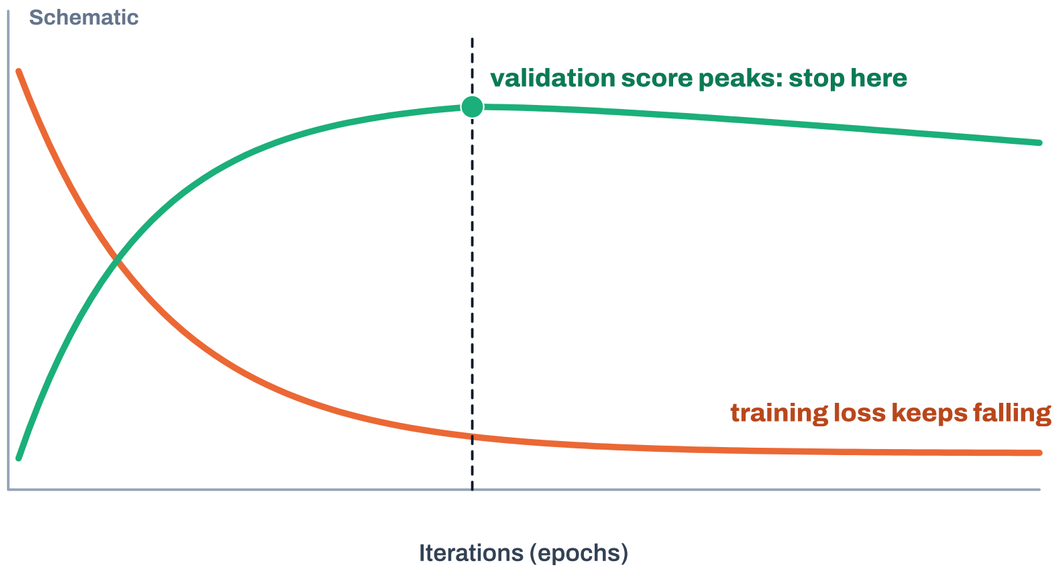
Tracks the optimisation objective on training batches. It is not the test RMSE.

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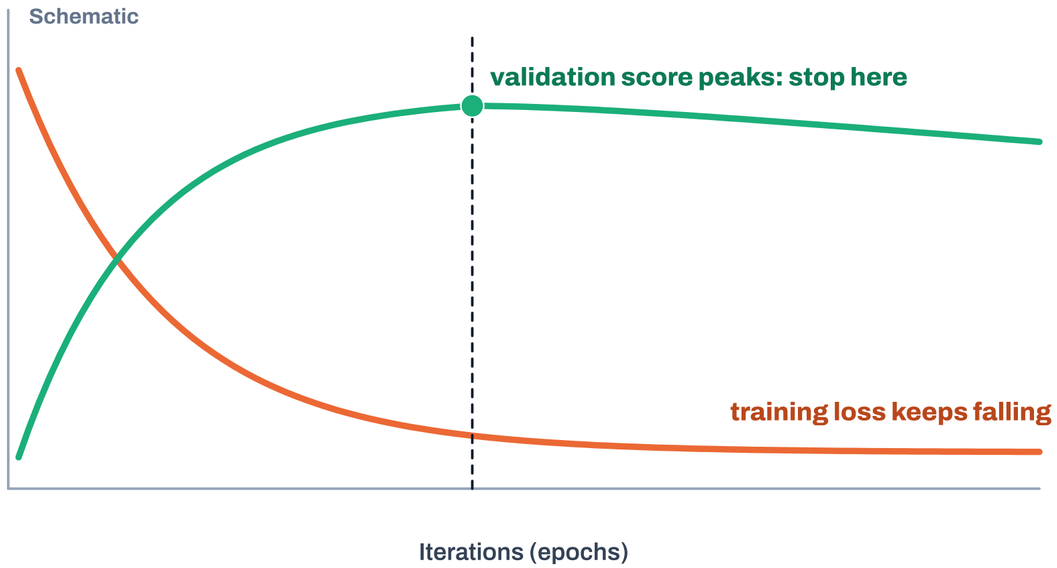
Performance on a held-out part of the training data, used by early stopping.

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INTERNAL VALIDATION R^2

Performance on a held-out part of the training data, used by early stopping.

A falling training loss is not evidence of generalisation.

Inspect both curves; note if training hit the iteration limit.

The internal validation subset is not the final test set.

THREE DATA ROLES · DO NOT MIX THEM UP

Train, validate, then test once



TRAINING DATA

Fit the parameters

Weights and biases are learned here.

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VALIDATION DATA

Monitor and choose

Early stopping, and comparing candidate architectures.

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Used once, after every choice is made.

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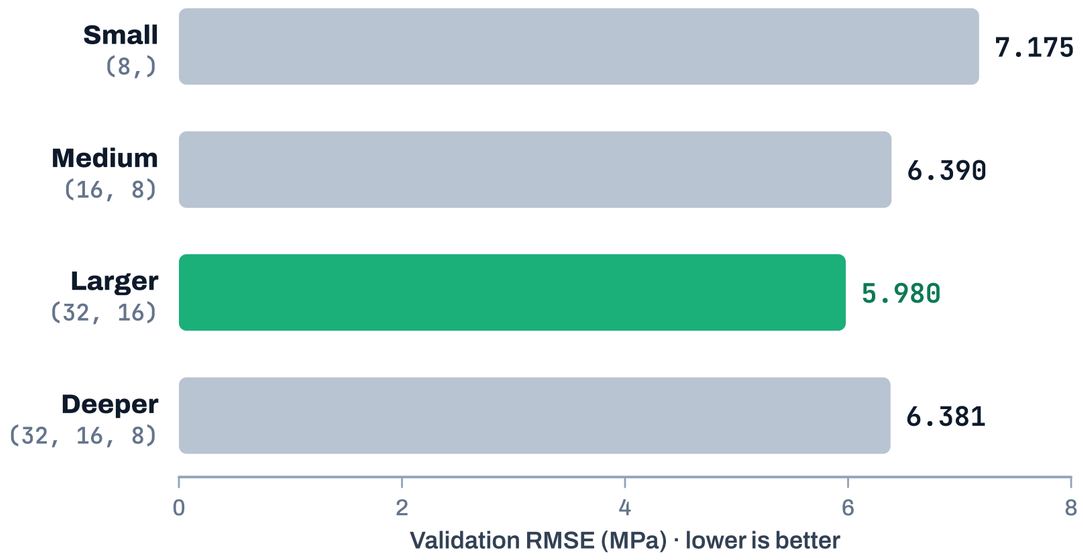
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The architecture is chosen on validation RMSE. The held-out test set plays no part in that choice.

CANDIDATE ARCHITECTURES, COMPARED ON VALIDATION DATA

Does a bigger ANN automatically do better?

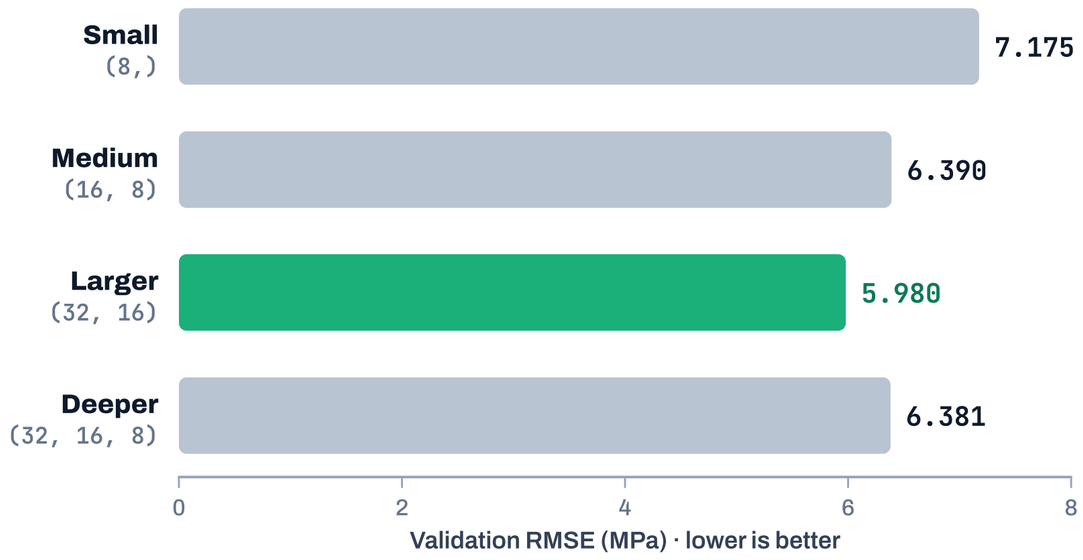
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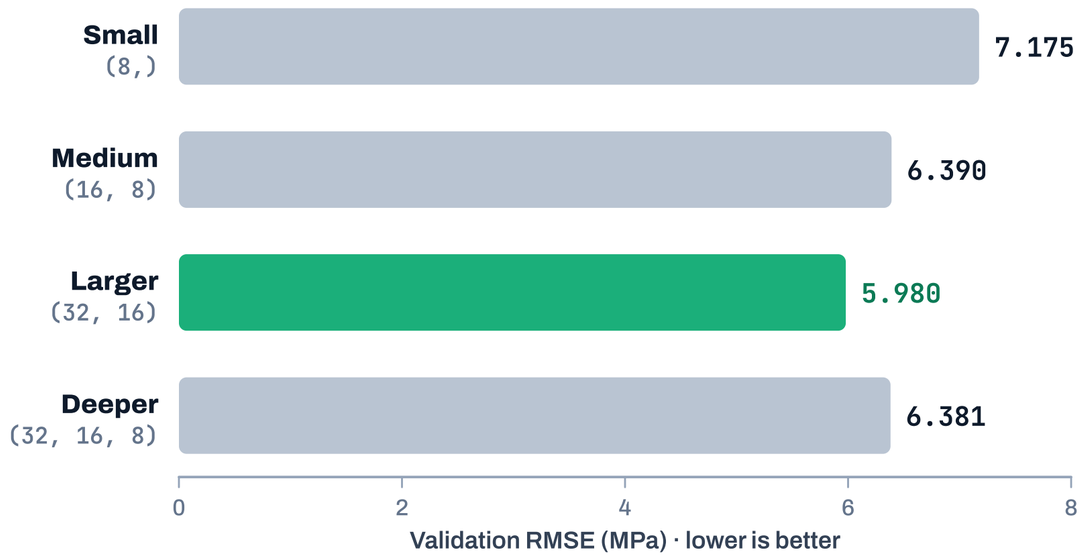
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Lowest validation RMSE of the four candidates: **5.980 MPa**.

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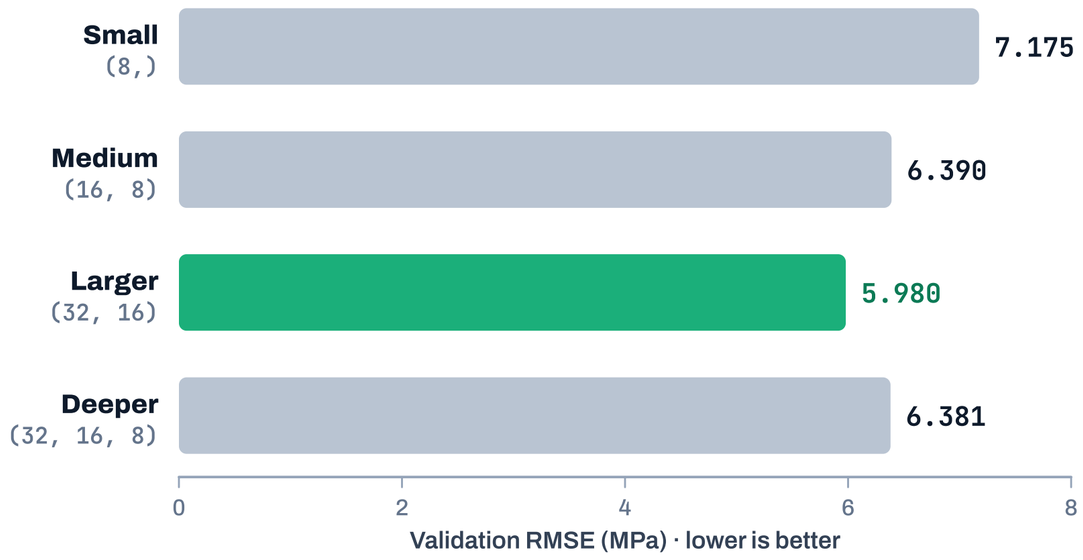
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Adding a third layer, (32, 16, 8), gave 6.381 MPa: no improvement.

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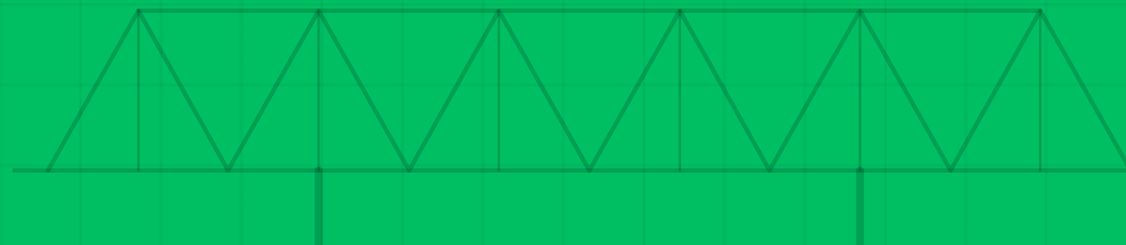
Best of four tried. Not proof of a global optimum.

03

- a Measuring regression performance
- b Results on the held-out test set
- c What R^2 does and does not tell us
- d Actual vs predicted, and residuals
- e Does an ANN understand concrete?
- f Transfer to geotechnical problems

PART 03 · RESULTS & JUDGEMENT

Is the ANN worth its complexity?



HOW DO WE MEASURE REGRESSION PERFORMANCE?

Complementary metrics, not one number

MEAN ABSOLUTE ERROR

MAE

$$\text{mean } |y - \hat{y}|$$

The average size of an error, in MPa.

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$$\sqrt{\text{mean } (y - \hat{y})^2}$$

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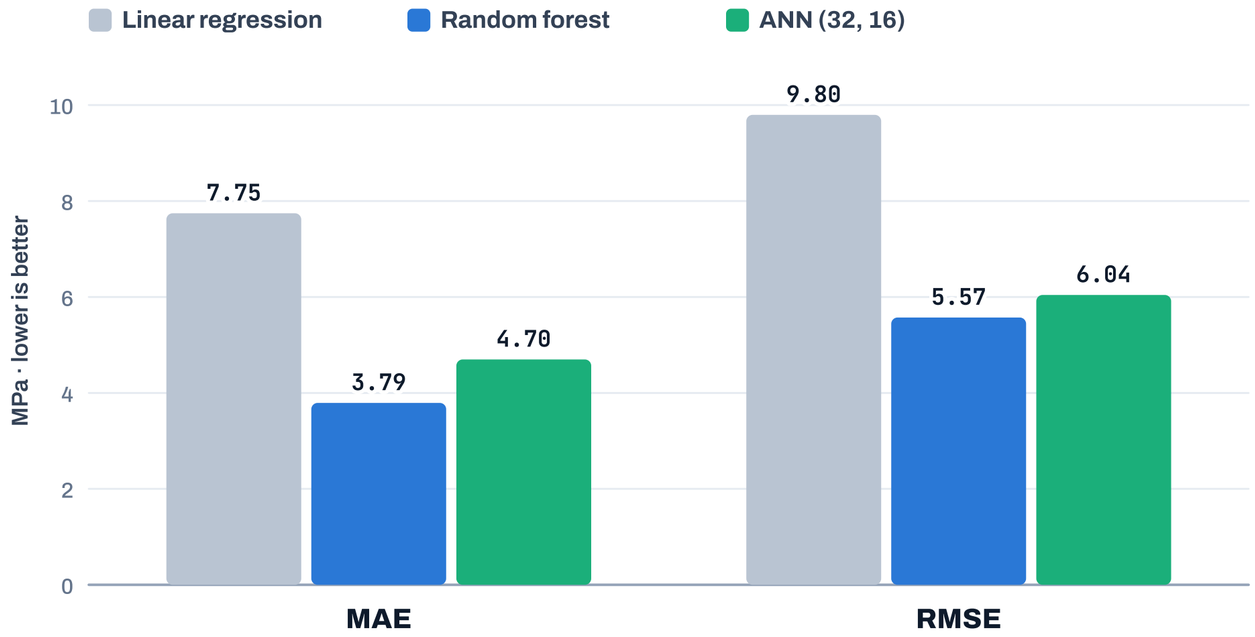
$e = y - \hat{y}$ **Residual > 0**: actual above predicted, the model underpredicts.

Residual < 0: actual below predicted, the model overpredicts.

ONE FIXED 80/20 SPLIT · THE SAME 206 TEST MIXES FOR ALL THREE MODELS

Results on the held-out test set

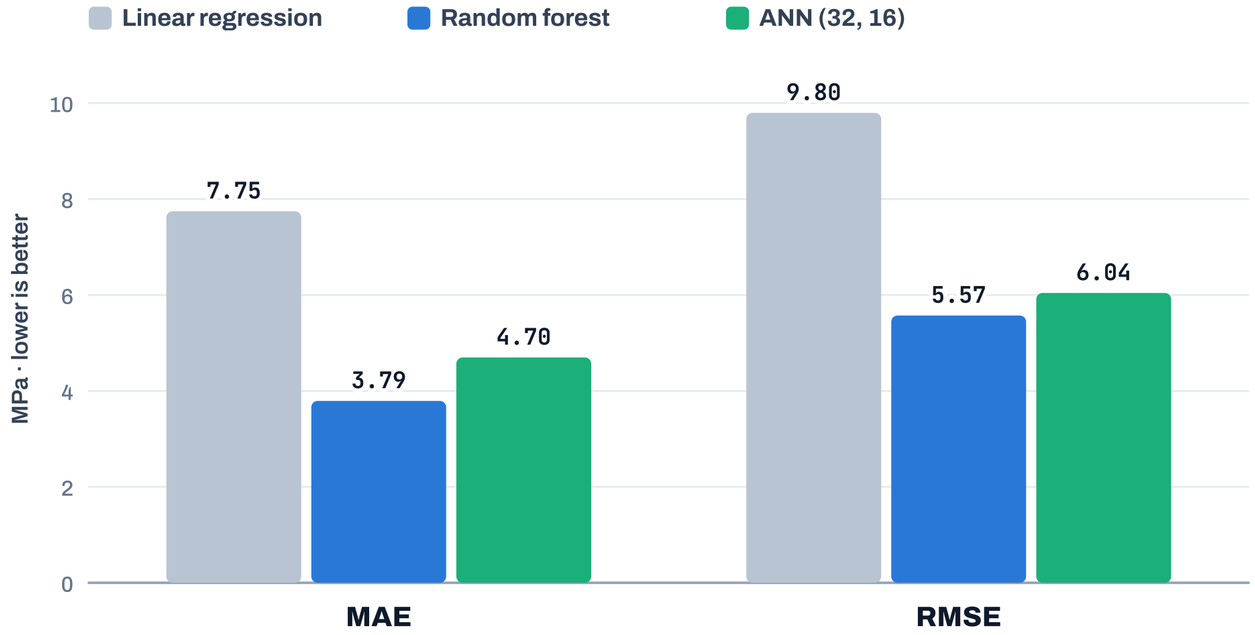
Test errors



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RANDOM FOREST

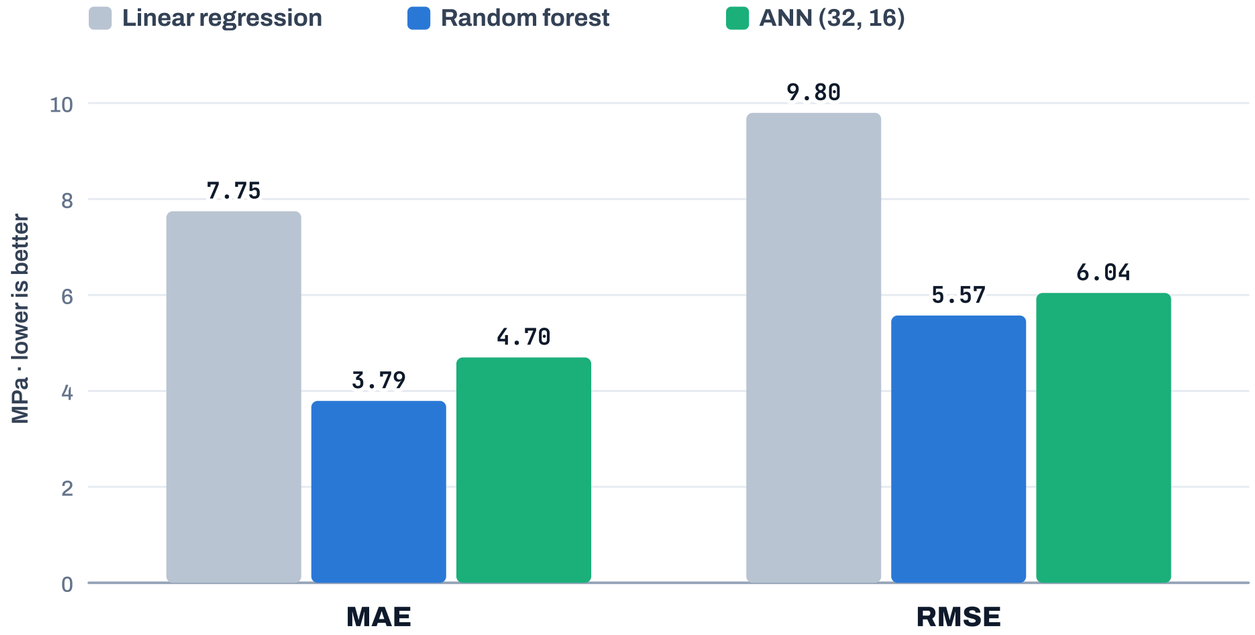
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Best of these configurations in this run.

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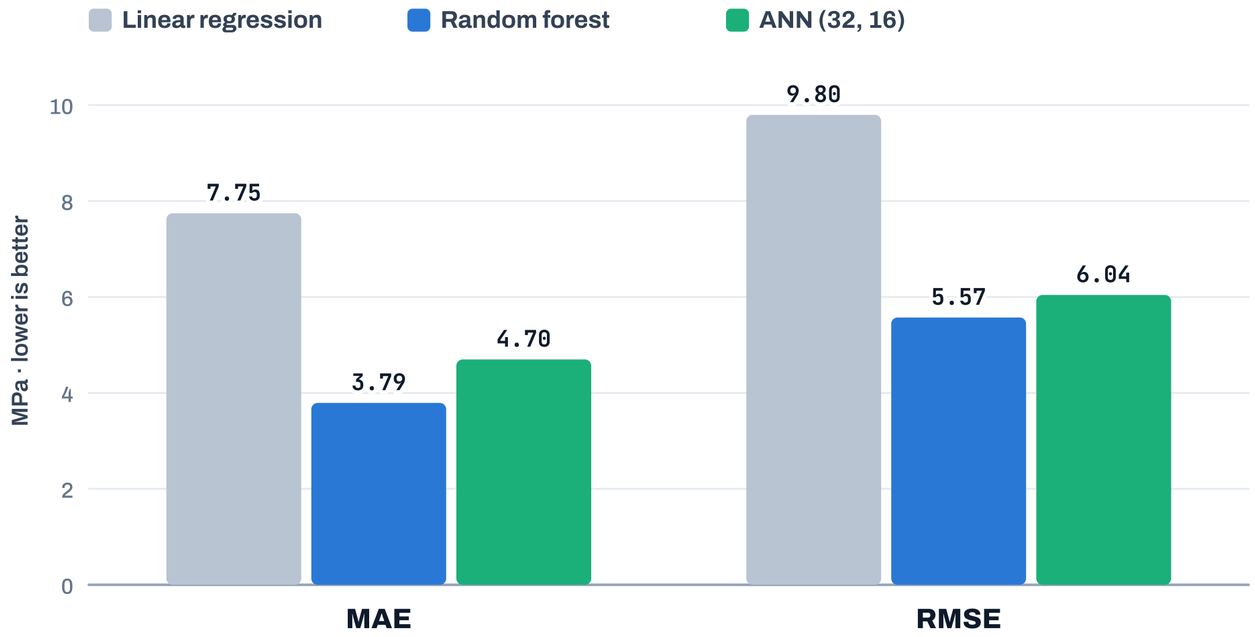
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Competitive, but not the best.

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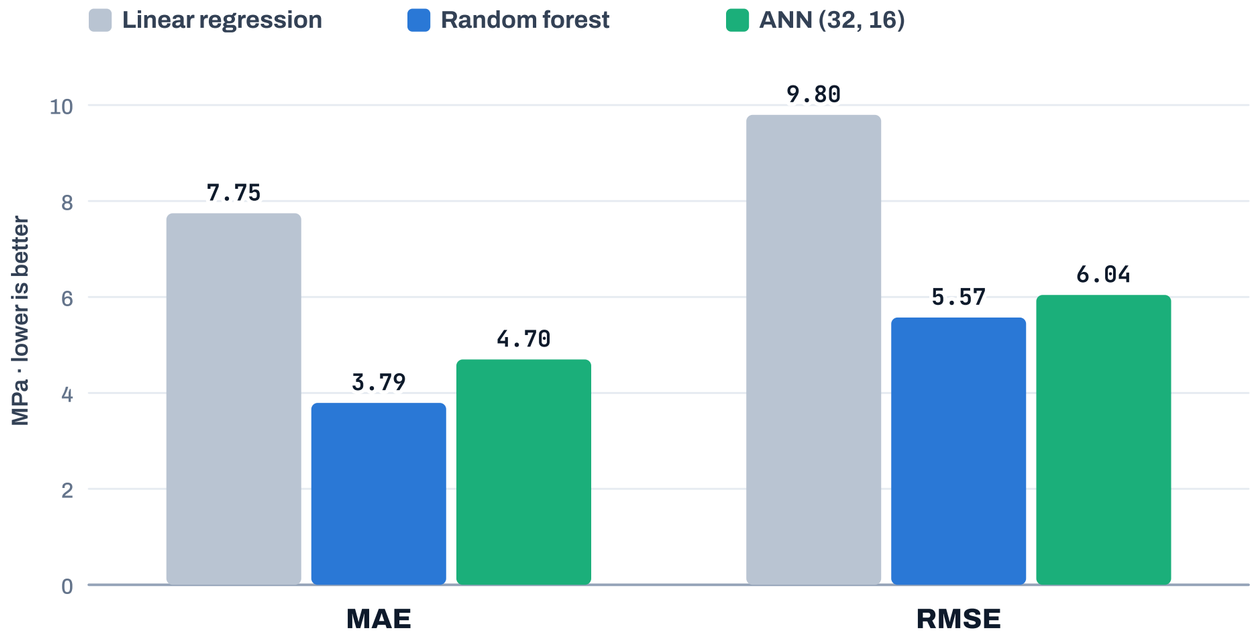
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Both nonlinear models clearly improve on it.

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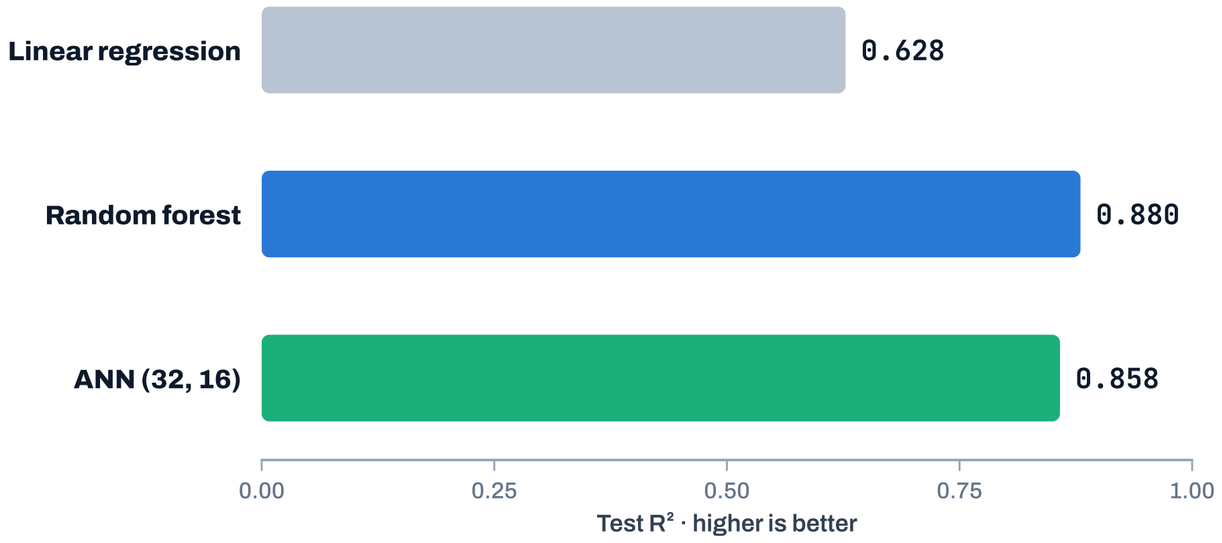
Both nonlinear models clearly improve on it.

One split, one run. These results do not prove which model family is best in general.

WHAT DOES R^2 TELL US HERE?

Useful, but not a complete measure of quality

Share of strength variance explained on the test mixes



BEST OBSERVED R^2

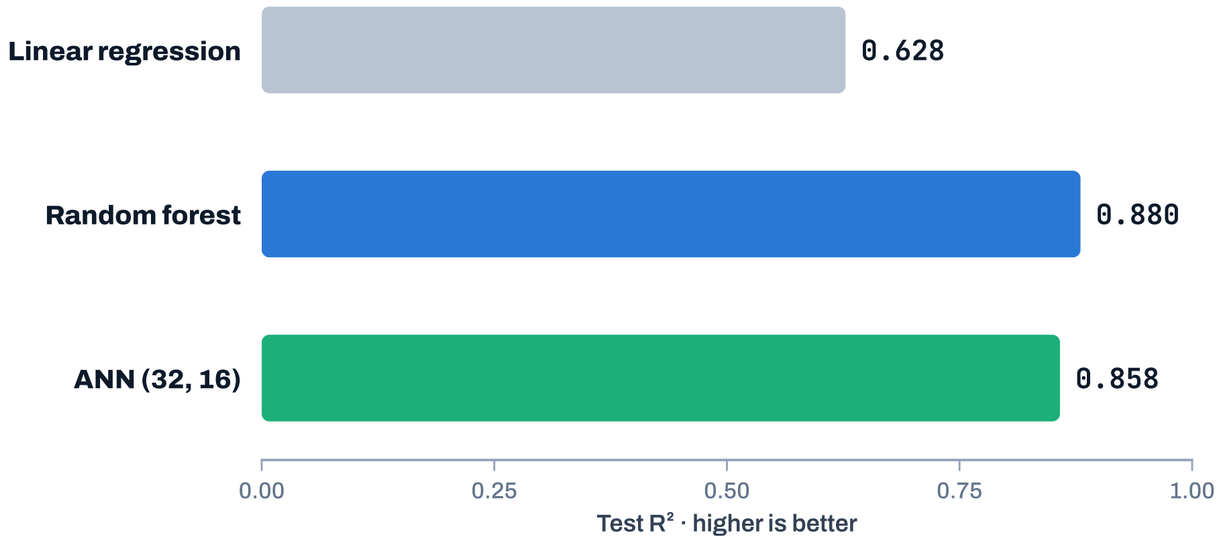
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Random forest; the ANN reaches 0.858.

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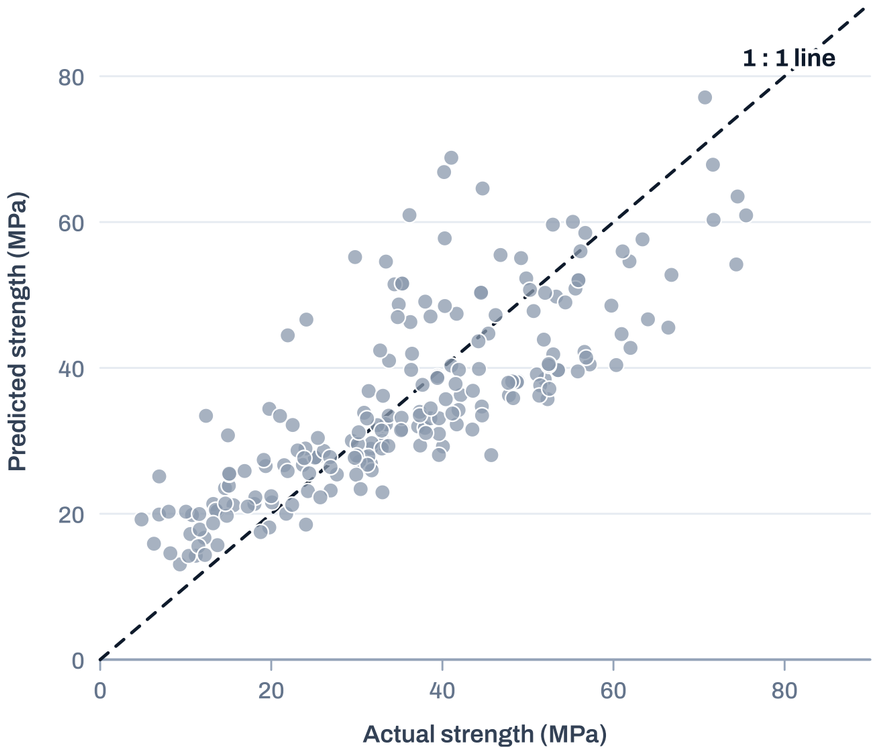
R^2 ALONE DOES NOT SHOW

- Error patterns
- Performance near critical thresholds
- Validity outside the training domain

ACTUAL VERSUS PREDICTED · INSPECT THE PATTERN

Points near the dashed 1 : 1 line are accurate

Linear-regression baseline, 206 test mixes



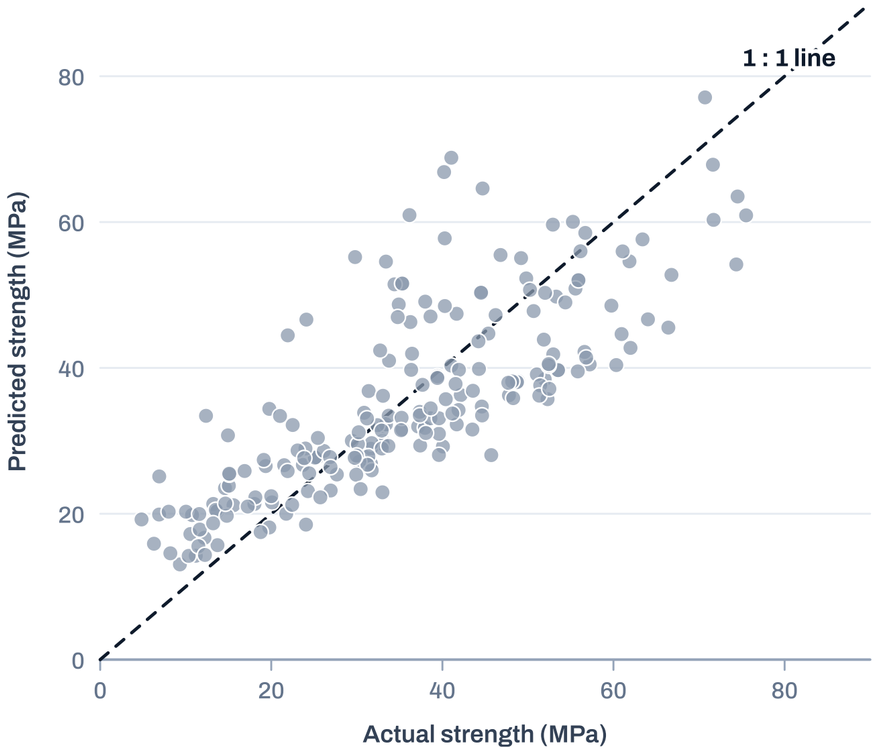
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- Scatter around the 1 : 1 line
- Systematic under- or overprediction
- Different spread at low and high strength

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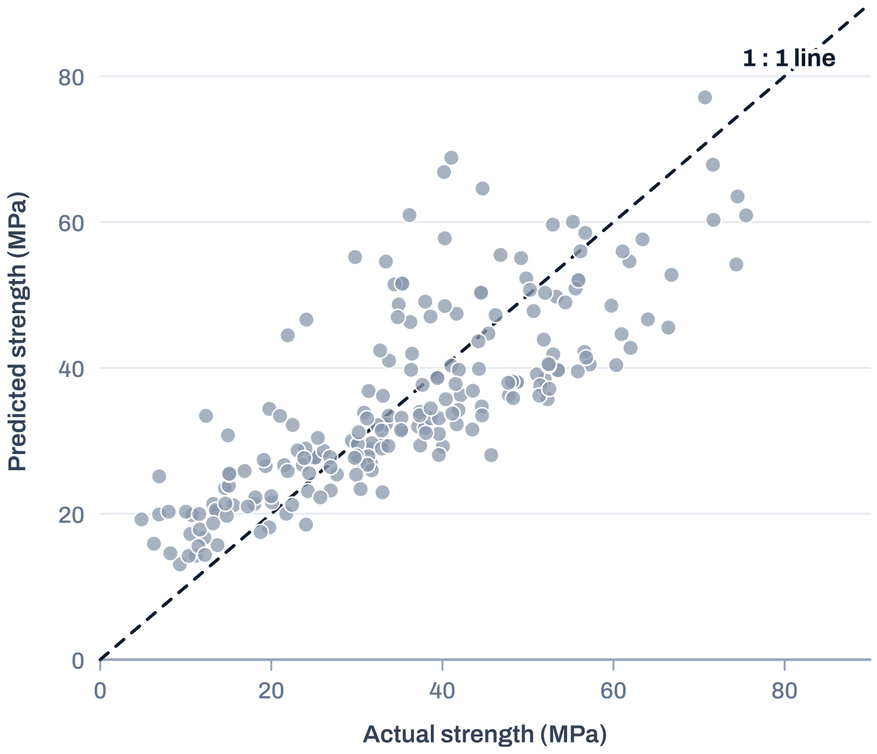
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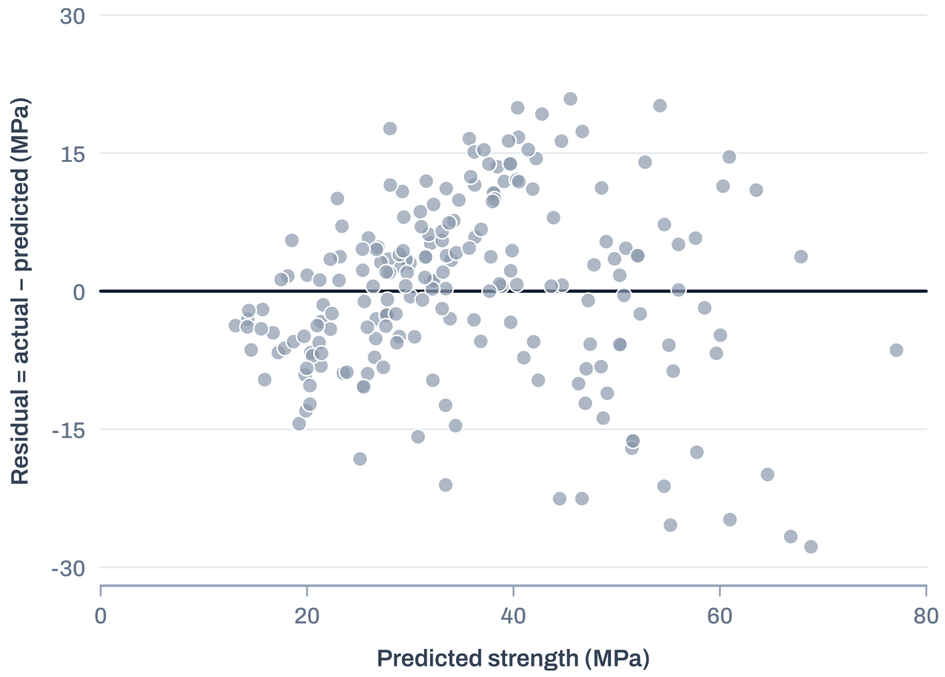
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Live: the ANN and random-forest plots in the notebook.

RESIDUALS · ACTUAL MINUS PREDICTED, FOR EVERY TEST MIX

Residuals reveal systematic errors

Linear-regression residuals, test mixes



POSITIVE RESIDUAL
Actual strength above predicted: underprediction.

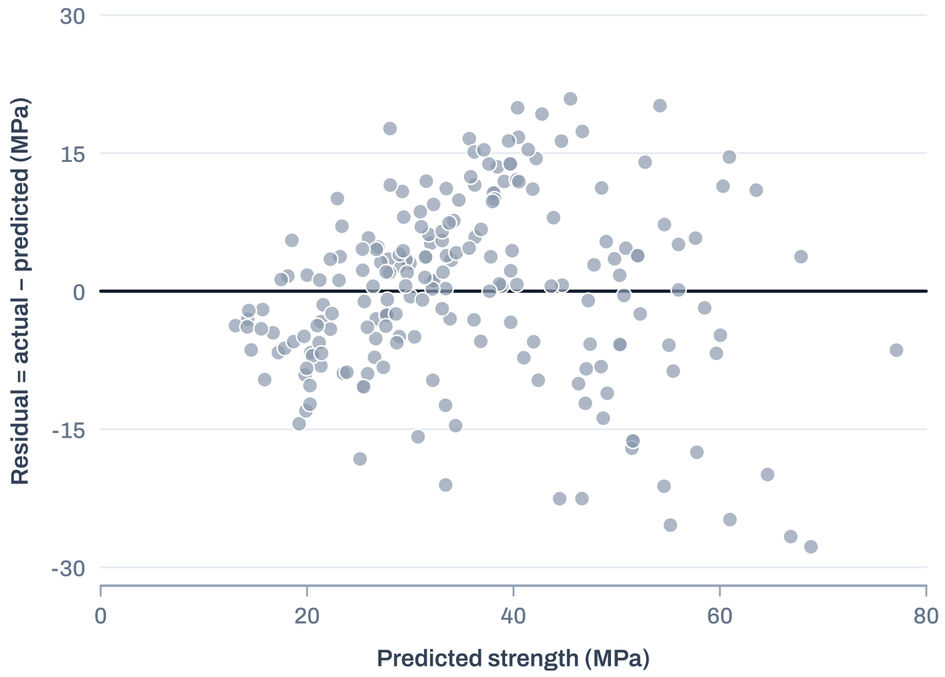
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May show systematic error, or unequal error spread.

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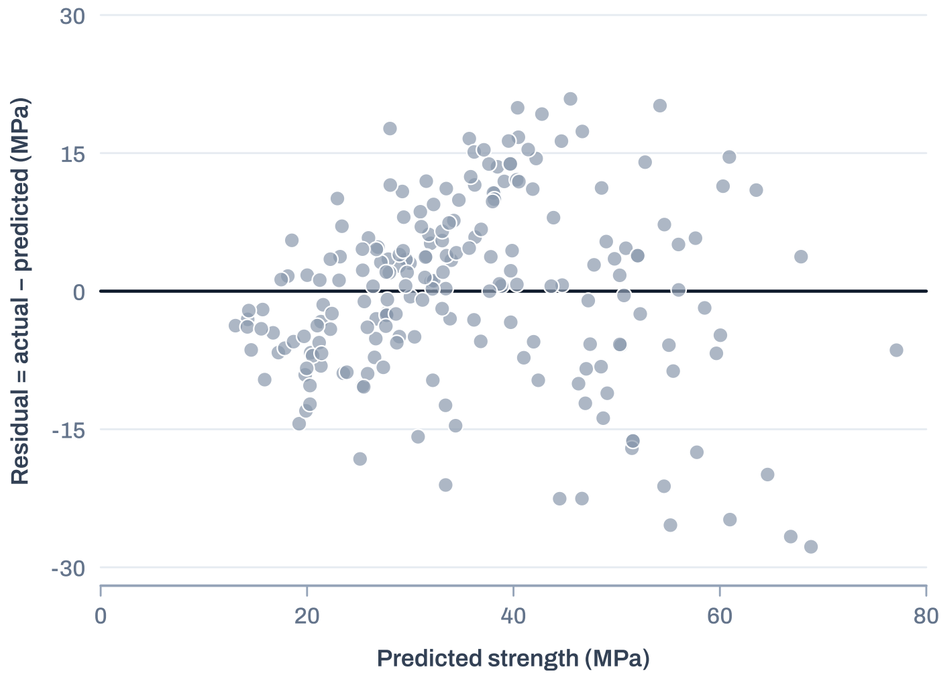
IN THIS BASELINE

Weak mixes (actual < 20 MPa) are overpredicted by **7.5** MPa on average; strong mixes (≥ 60 MPa) underpredicted by **12.0** MPa.

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⚠ A good average score can hide weak performance in parts of the strength range.

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No: it learns a statistical mapping from examples

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THE TRAINING DOMAIN

Cement	102–540 kg/m³
Water	122–247 kg/m³
Superplasticizer	0.0–32.2 kg/m³
Age	1–365 days
Strength	2.3–82.6 MPa

A mix outside these ranges, or an unusual combination inside them, is extrapolation.

ILLUSTRATIVE APPLICATIONS · NOT TRAINED OR VALIDATED MODELS

The same idea transfers to geotechnical engineering

Slope-stability surrogate

Geometry + soil parameters + hydraulic conditions

→ approximate factor of safety

Like the 616 Morgenstern–Price slopes of Session 05, as a regression on FOS.

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The basis of the ML surrogates in Session 08.

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A surrogate speeds up repeated predictions. It does not validate the underlying numerical model or justify extrapolation.

TAKE-HOME MESSAGES

ANNs are useful tools, not automatic upgrades

1

Understand the mapping

Inputs → weighted combinations → nonlinear activations → output.

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Loss, gradients and parameter updates shape the learned model.

3

Validate correctly

Validation data for choices; the test set only for the final evaluation.

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Here, the random forest beat the selected ANN on the held-out test set.

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Keep claims proportional

Scores do not establish causation, physical validity or safe extrapolation.

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THE CENTRAL LESSON

Choose the model the evidence supports, not the most sophisticated one.

AFTER THE SESSION · USE THE NOTEBOOK TO REINFORCE THE WORKFLOW

Five things to do in the notebook

01	Compare the three fitted models using MAE, RMSE and R^2
02	Inspect the actual-versus-predicted plots and the ANN residual plot
03	Explain why (32, 16) was selected using validation RMSE
04	Discuss whether a more complex model is justified by its performance
05	Name two limitations before considering real engineering use

SESSION 06 NOTEBOOK

Link shared in the session chat

Dataset: Yeh, I.-C. (1998). Concrete Compressive Strength. UCI Machine Learning Repository. doi:10.24432/C5PK67

Questions

Unmute, or drop your question in the chat.



Neurons & layers

Activations

Training & validation

ANN vs baselines

Dr. Abdul Waris
Research Associate · Universität Siegen, Germany

END OF SESSION 06

Thank you.

See you in session 07: **Can We Trust Our ML Model?**

Dr. Abdul Waris
Research Associate · Universität Siegen, Germany

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